



VILNIUS GEDIMINAS TECHNICAL UNIVERSITY
FACULTY OF MECHANICS
DEPARTMENT OF MECHATRONICS, ROBOTICS AND DIGITAL MANUFACTURING

Baha Celik

**Integration of Computer Vision and Natural Language Processing for
Autonomous Solar Panel Defect Detection and Classification**
Final Bachelor's Project

Study programme APPLIED ARTIFICIAL INTELLIGENCE,
Code 6121EX084

Vilnius, 2026

VILNIUS GEDIMINAS TECHNICAL UNIVERSITY
FACULTY OF MECHANICS
DEPARTMENT OF MECHATRONICS, ROBOTICS AND DIGITAL MANUFACTURING

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Head of Department

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Supervisor: Mantas Makulavičius _____
(Title, Name, Surname) (Signature) (Date)

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DEPARTMENT OF MECHATRONICS, ROBOTICS AND DIGITAL MANUFACTURING

Production and Manufacturing Engineering study field
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APPROVED BY
Head of Department
Vytautas Bučinskas
2026-05-08

OBJECTIVES FOR BACHELOR'S DEGREE FINAL WORK (PROJECT)

No. TDIfu-22-12165

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For student Baha Celik

Final work (project) title: Integration of Computer Vision and Natural Language Processing for Autonomous Solar Panel Defect Detection and Classification

Deadline for completion of the final work (project) according to the planned study schedule.

THE OBJECTIVES:

: The total mass of the autonomous defect detection unit, including the single board computer, camera assembly, and casing must not exceed 1.5 kg; the physical dimensions of the enclosure must not exceed 200 mm × 150 mm × 150 mm to ensure compatibility with standard solar panel frames; the vision system must be capable of resolving surface defects (bird drop, dust, physical damage, snow-cover, etc.) with a minimum feature size of 1 mm from a distance of 1m; the onboard inference engine must process images with a latency of no more than 3 seconds per frame while maintaining a mean Average Precision (mAP) of at least 85%; the agentic reporting module must be capable of aggregating classification data, generating a natural language summary, and transmitting a formatted email report including the image within 60 seconds of inspection completion.

:

1. Introduction. Analysis of similar designs. Justification of decisions made.
2. Description of the system design and operating principle.
3. Description of the application of the artificial intelligence-based system.
4. Creation of a control system.
5. Establishment of requirements for safe operation and data security using the device.
6. Calculation of economic indicators.
7. Final conclusions.
8. List of references.

:

1. General drawing of the system (1 page A1);
2. Assembly drawing of the system (node) (1 page A1);
3. Diagram of the applied artificial intelligence architecture (0.5 pages A1);
4. Control algorithm of the designed system (0.5 pages A1);
5. Structural block diagram of the system (0.25 pages A1);
6. Work drawing of selected part (0.25 pages A1);
7. Economic indicators (0.5 pages A1).

Academic Supervisor Mantas Makulavičius

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Annotation This final bachelor's thesis presents an alternative solution for proactive maintenance of residential solar panels. By combining natural language processing and computer vision, it offers a competitive edge over existing market solutions. With cost efficiency prioritized, the smart trigger system conserves expensive resources. Component selection was optimized for edge computation, reliability, and market analysis. The system comprises a weatherproof, custom 3D enclosure printed with Acrylonitrile Styrene Acrylate; a Raspberry Pi 5 with 8 GB of RAM serving as the microcomputer; an Active Cooler for thermal management in outdoor settings; a Camera Module 3 as the visual sensor for the computer vision model; an AM2302/DHT22 temperature and humidity sensor for regional temperature and humidity metrics; and a power supply serving as the main power source. Mechanical and analytical calculations confirmed structural feasibility. The final system is estimated to weigh 290 g. The general dimensions of the system are 179 mm x 56 mm x 133 mm. An abstracted algorithmic flow and an AI architecture diagram were also created. Economic analysis revealed a bill of materials of 216 EUR, a direct variable manufacturing cost of 301 EUR, annual fixed costs of 6000 EUR, a commercial break-even point of 40.3 units, and a payback point of 5.25 years. With an ROI of 19.05%, these economic indicators confirm the commercial viability of this highly efficient modern concept for solar panel maintenance and defect detection.			
Keywords: Computer Vision, Natural Language Processing, Raspberry Pi, Edge Computation, Solar Panel, Maintenance, Defect, Detection, Large Language Models, Visual Language Models, Agents, Artificial Intelligence.			

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Pirmosios pakopos studijų Taikomojo dirbtinio intelekto programos bakalauro baigiamasis darbas 3	
Pavadinimas	Kompiuterinio matymo ir natūralios kalbos apdorojimo integravimas autonominiam saulės kolektorių defektų aptikimui ir klasifikavimui
Autorius	Baha Celik
Vadovas	Mantas Makulavičius

	Kalba: anglų
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Anotacija

Šiame baigiamajame bakalauro darbe pristatomas alternatyvus gyvenamųjų namų saulės modulių proaktyvios priežiūros sprendimas. Derinant natūralios kalbos apdorojimą ir kompiuterinę regą, siūlomas sprendimas suteikia konkurencinį pranašumą prieš esamus rinkos sprendimus. Kadangi prioritetas teikiamas kaštų efektyvumui, išmanioji paleidimo sistema padeda taupyti brangius išteklius. Komponentų parinkimas buvo optimizuotas atsižvelgiant į kraštinių skaičiavimų galimybes, patikimumą ir rinkos analizę. Sistemą sudaro oro sąlygoms atsparus individualiai suprojektuotas 3D spausdintas korpusas, pagamintas iš akrilnitrilo stireno akrilato; „Raspberry Pi 5“ su 8 GB RAM, veikiantis kaip mikrokompiuteris; aktyvus aušintuvas, skirtas šilumos valdymui lauko sąlygomis; „Camera Module 3“, naudojamas kaip kompiuterinės regos modelio vaizdo jutiklis; AM2302/DHT22 temperatūros ir drėgmės jutiklis, skirtas regioniniams temperatūros ir drėgmės rodikliams matuoti; bei maitinimo šaltinis, veikiantis kaip pagrindinis energijos šaltinis.

Mechaniniai ir analitiniai skaičiavimai patvirtino konstrukcijos įgyvendinamumą. Galutinės sistemos numatoma masė yra 290 g. Bendrieji sistemos matmenys yra 179 mm × 56 mm × 133 mm. Taip pat buvo sudaryta apibendrinta algoritminė veikimo schema ir DI architektūros diagrama. Ekonominė analizė parodė, kad medžiagų sąmata siekia 216 EUR, tiesioginės kintamosios gamybos sąnaudos – 301 EUR, metinės pastoviosios sąnaudos – 6000 EUR, komercinis lūžio taškas – 40,3 vieneto, o atsipirkimo laikotarpis – 5,25 metų. Esant 19,05 % investicijų grąžai, šie ekonominiai rodikliai patvirtina šios itin efektyvios modernios saulės modulių priežiūros ir defektų aptikimo koncepcijos komercinį gyvybingumą.

Prasminiai žodžiai: Kompiuterinė rega, natūralios kalbos apdorojimas, „Raspberry Pi“, kraštiniai skaičiavimai, saulės modulis, priežiūra, defektai, aptikimas, didieji kalbos modeliai, vaizdo ir kalbos modeliai, agentai, dirbtinis intelektas.

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INTRODUCTION

Investigation problem. Global transition towards environmentally friendly options, such as renewable energy, has led to an immense rise in photovoltaic sources. Moreover, this big shift comes with its maintenance and operational demands. Current demand for large-scale maintenance solutions is the norm. However, these photovoltaic sources are becoming more accessible every day and are being preferred for small-scale residential settings, leaving residential owners to rely on reactive and manual maintenance. Therefore, there is a critical gap in maintenance technologies in the sector, often referred to as “scale-bias”. This neglect has significant economic and energetic consequences. For instance, research shows that unchecked soiling can reduce annual energy output by 5% under normal conditions (Photovoltaic Module Soiling Map | Photovoltaic Research | NLR, n.d.). In comparison, other naturally occurring issues, such as bird droppings, can lead to steeper losses of 20% to 50% (Maghami et al., 2016). Consequently, this thesis addresses the lack of an autonomous, cost-effective, and pragmatic maintenance solution specifically designed for small-scale photovoltaic installations.

Object of project. To address this gap, the objective is to develop an autonomous monitoring system for the maintenance of residential solar arrays, with the help of an efficient and modern single-board computer, such as the Raspberry Pi 5, and a high-resolution Camera Module 3 housed in a custom enclosure that enables a compact system. The wide-angle model of the high-resolution camera gives the system the best possible angle, making the proposed system a great fit for the industry problem. This hardware is aided by a complex software architecture that combines lightweight edge computing, real-time computer vision for defect detection, You Only Look Once (YOLO), with a state-of-the-art Agentic AI framework that utilizes large language models for high-level decision making and action report creation (CrewAI).

The aims and purposes of the project. The primary goal of this system is to offer a utilitarian solution for small-scale residential solar panel setup by a scan and sleep method for power efficiency and autonomous analysis for better user experience.

The novelty and relevance of the theme. As a result, application of Agentic AI within an IoT context, unlike traditional systems that strictly focus on binary detection, introduces a semantic and deeper layer of understanding, enabling the system to reason about environmental context and other factors before recommending actions for people who do not have or do not want to give more effort and time for their photovoltaic systems. This approach is advantageous to other alternatives, such as manual checkups or large-scale drone and robotic services, which are extremely costly and impractical. By shifting the delivery model from an expensive service to an affordable product, this work presents a relevant and scalable solution to the market gap of the solar maintenance market, ensuring that the efficiency gains of modern technology are accessible to the mass market.

1. OVERVIEW OF ANALOGIC CONSTRUCTIONS AND SPECIFIC NODES

1.1. Existing solutions

The industry of solar panel maintenance and defect detection is mainly addressed in the context of large photovoltaic (PV) plants. Therefore, the market is saturated with dynamic, high-scalability robotics solutions designed to inspect megawatts of capacity efficiently (Meribout et al., 2023). For instance, the largest power plant in Europe, the Karapinar solar power plant, located in Turkey, houses approximately 3.5 million solar panels across an area of 20 million square meters with a peak capacity of 1.35 Gigawatts (GW) of power while also preventing 1.7 million tons of carbon emissions annually (Kalyon Holding, n.d.). As seen in Fig. 1.1 below, the endless solar panels require a highly scalable solution to maintain the entire power plant, as it is virtually impossible to maintain millions of solar panels manually.



Fig. 1.1. Kalyon Holdings “Karapinar” solar power plant (Kalyon Holding, n.d.)

These large-scale solutions generally fall under aerial, ground-based, and maintenance-action categories. It is also worth mentioning autonomous swarm solutions that can be applicable for each of the categories defined above as an additional feature. While specialized products for small-scale applications exist, the underlying defect detection methods, specifically visual and thermal imaging, remain consistent across all scales.

1.1.1. Aerial solutions

Unmanned Aerial Vehicles (UAVs) represent the industry standard for rapid, large-scale inspection. Companies such as “Raptor Maps” (Raptor Maps, n.d.) and “DroneDeploy” (*Drone Solar Inspection | Renewable Energy | DroneDeploy*, n.d.-a) provide enterprise-grade software platforms that ingest data captured by drones to generate orthomosaic and thermal maps. These solutions leverage thermography to detect hotspots and string outages from an altitude of 20–50 meters. While highly effective for covering large areas, aerial systems face limitations regarding battery flight time, regulatory airspace restrictions, and the inability to capture close-range, high-fidelity surface details (e.g., micro-cracks or specific soiling patterns) without flying dangerously close to panels.

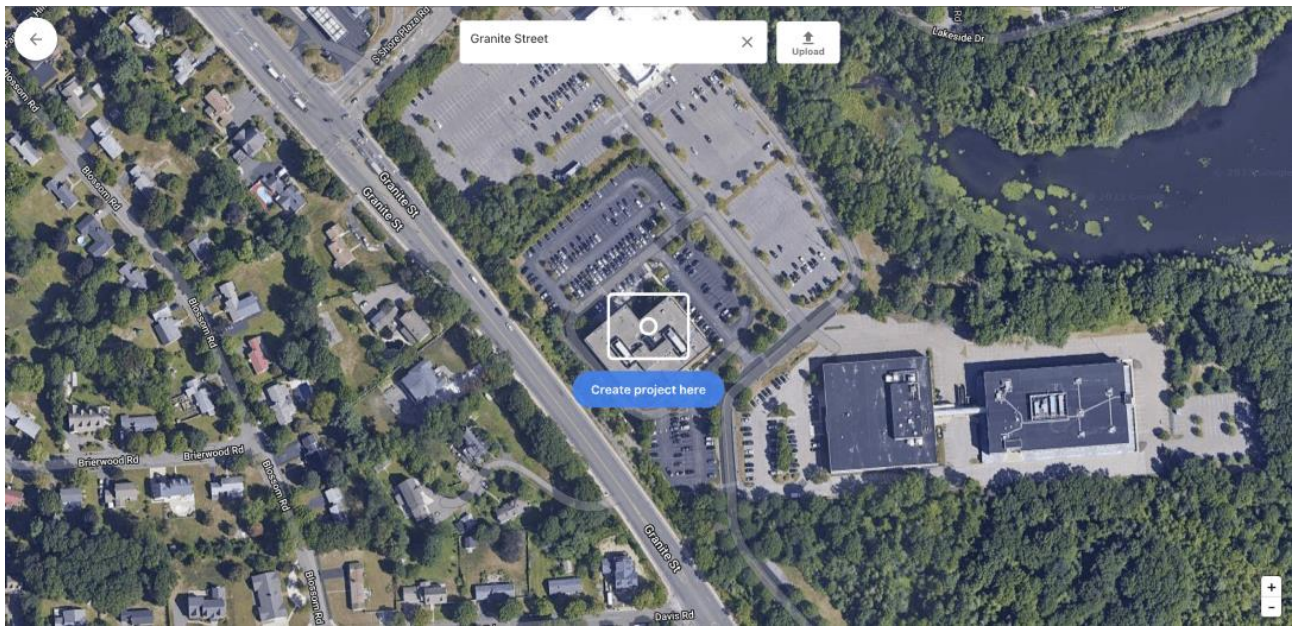


Fig. 1.2 Drone Deploy Aerial Maintenance setup software (DroneDeploy, n.d.)

Setting up an area for maintenance in rooftop conditions could also endanger human lives; companies like Drone Deploy offer a fully autonomous method to eliminate the occupational hazard by using drones through a map-like interface to create a job for the mentioned drone to automatically maintain an area and map it for a better user experience as seen in Fig. 1.2 (*Drone Solar Inspection | Renewable Energy | DroneDeploy*, n.d.-b). Users can then act on the insights provided by these inspection systems, making the procedure efficient and safe. The same software could also be used to plan future projects, gaining as realistic information about the terrain and its conditions as possible.

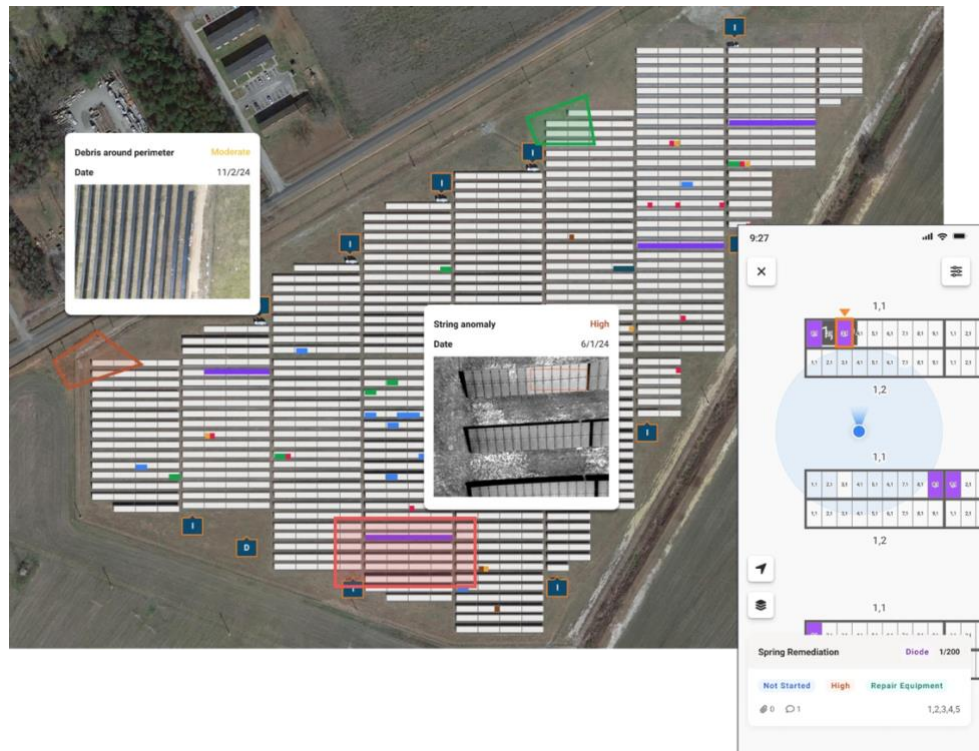


Fig. 1.3. Raptor maps software aerial view (Raptor Maps, n.d.)

The aerial view in the Raptor Maps software in Fig. 1.3 provides a clear overview of the situation. It even divides the power plant into sectors to easily pinpoint and address the issue (*Solar's Most Comprehensive Inspections Suite | Raptor Maps, n.d.*).

1.1.2. Ground robotics solutions

For more granular inspection, Unmanned Ground Vehicles (UGVs) offer a stable alternative that can operate for longer durations than drones, due to fewer weight and shape constraints. A leading example is “OnSight Technology,” which uses the "Bulldog" autonomous rover (*Ground Inspection Robot – OnSight, n.d.*). Unlike drones, these rovers drive between panel rows, using gimbal-mounted thermal and optical cameras to inspect both the front-facing glass and the rear side of these modules.



Fig. 1.4 OnSight technology bulldog robot (*Ground Inspection Robot – OnSight, n.d.*)

As can be seen in Fig. 1.4, the robot's height and maneuverability provide advantages in accuracy and the ability to perform a more thorough analysis. It goes at 1.6 kilometers per hour for up to 12 hours a day on a single charge. The robot can inspect up to 580 kilometers, equivalent to approximately 7.5 technicians' labor (*Ground Inspection Robot – Onsight*, n.d.). These systems provide higher stability and the ability to perform "under-panel" diagnostics. However, they require significant capital investment and flat, navigable terrain, making them unsuitable for roof-mounted or smaller, irregular installations.

1.1.3. Autonomous and swarm solutions

To manage swarm operations, Companies like “vHive” utilize "vHive Capture" and “vHive Analytics” technology to coordinate swarms of autonomous drones (*Autonomous Solar Farms Inspection Software | VHive*, n.d.). These platforms allow multiple agents to inspect a site simultaneously, stitching data together into a cohesive 3D model. Utilizing multiple drones or robots significantly cuts the duration it takes to scan a field and makes the process more efficient as observed in Fig. 1.5. While vHive demonstrates the power of multi-agent orchestration (like the *Agentic AI* workflow proposed in this thesis), it relies on expensive commercial drone hardware and proprietary cloud processing, creating a high barrier to entry for smaller operators.



Fig. 1.5. vHive field intelligence done by autonomous drones (*VHive Capture | VHive*, n.d.)

By using AI-driven anomaly detection that identifies thermal irregularities like hotspots, bypass diode failures, and potential induced degradation (PID), these systems allow for automated classification that meets IEC TS 62446-3 standards for outdoor thermography (IEC, 2017). Using multiple agents greatly reduces data acquisition time through linear scalability. This effectively divides the scan duration by the number of active drones, while still maintaining high Ground Sampling Distances (GSD) for sub-centimeter precision, as shown in Fig. 1.5. vHive shows the benefits of multi-agent coordination.

1.1.4. Maintenance and action solutions

While the above solutions focus on *detection*, other systems focus on remediation. Companies like “SolarCleano”(SolarCleano, n.d.) and “HyCleaner”(Cleaning Solutions for Solar, Facade & Glass, n.d.) manufacture modular robots that physically attach to and drive across solar arrays to clean panel surfaces.



Fig. 1.6. SolarCleano M1 cleaning and maintaining a solar panel (*Home | SolarCleano, n.d.*)

While primarily designed for cleaning (removing soiling to restore efficiency), as seen in Fig. 1.6, modern iterations of these robots are increasingly integrating cameras and sensors to perform simultaneous visual inspection. However, these are "contact" devices. Heavy, mechanically complex, and carrying a risk of physical damage to the panels if a malfunction occurs.



Fig. 1.7 “solarROBOT pro” by HyCleaner operating on a rooftop (HyCLEANER, n.d.)

Another top player in the market, HyCleaner, with its “solarROBOT pro,” as shown in Fig. 1.7, can potentially save lives while operating in potentially dangerous conditions such as rooftops. The robot can clean up to 2400m² of solar panels in an hour. With 3 hours of battery life and a path-planning algorithm, it can operate without constant human assistance.

1.2. Computer vision architectures

Detection and classification are classical computer vision tasks that have been extensively researched and improved over the last few years. With these improvements, there is a range of unified models that could be used across various circumstances, from devices as small as a Raspberry Pi to a massive AI data centre in the cloud. Using these unified models often makes sense for numerous reasons, such as cost-effectiveness and performance, given the massive datasets they have been trained on. This gives a huge advantage over starting with a Convolutional Neural Network (CNN) architecture with random weights (Hidaka & Kurita, 2017).

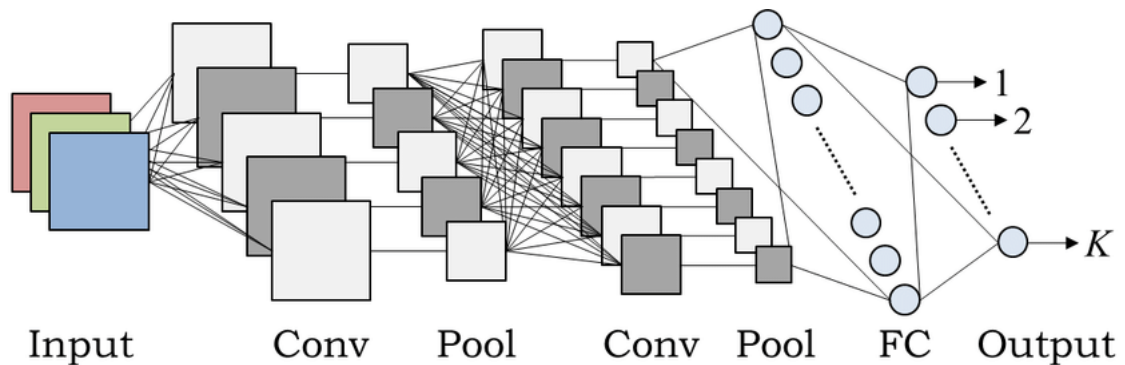


Fig. 1.8. Simple Convolutional Neural Network architecture with 5 layers (Raman et al., 2019)

Fig. 1.8 displays a simple convolutional neural network with multiple layers. The weights mentioned before are given to each node within a layer. This could be thought of as a relationship between a node in layer A and another node in layer B, or, mathematically, it determines how much an input from one node affects the output of the next. Due to their nature, neural networks get better at solving complex problems the bigger and more complex they are (Raman et al., 2019). In most cases, these neural networks are made for specific use cases that have meaningful connections only within that use case. Although this approach is more efficient for solving the problem at hand, it has architectural limitations in scalability and rigidity when the problem parameters change in the future (Sar et al., n.d.).

1.2.1. Classical computer vision

Before the widespread usage of deep learning, defect detection and classification relied on manual feature extraction and heuristic algorithms. They typically utilize edge detection techniques such as the Canny operator or the Sobel filters to differentiate the physical boundaries of photovoltaic modules, separating the active cell area from the background. Additionally, researchers often applied “feature engineering” using algorithms such as the Hough Transform to detect straight lines (e.g., panel frames) or Gabor filters to analyse surface textures for micro-cracks.

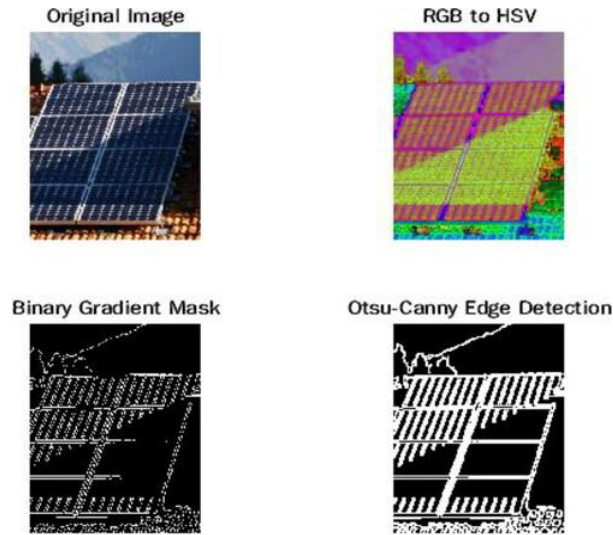


Fig. 1.9. Different filters applied to an image of a solar panel (Bommes et al., 2021)

For thermal use cases, statistical methods such as Otsu’s algorithm, as seen in Fig. 1.9, serve as the primary segmentation tool (Bommes et al., 2021). Converting an image to greyscale for binary classification to determine “hotspots” is a very computationally inexpensive and accurate technique, but it is limited by many bottlenecks, such as environmental noise, which is very frequent in real-world circumstances. These real-world scenarios could be as simple as a shadow cast by a nearby object or a patch of bird soil in our area of interest. Therefore, semantic understanding of the image is required for detection.

1.2.2. Two-Stage Deep Learning Detectors

To address the accuracy limitations of classical methods, a Two-Stage detector approach with a more complex “propose and verify” method is used. This paradigm would first detect potential regions of interest and then examine them to verify that they are the correct ones. Architectures such as “Faster R-CNN” and “Mask R-CNN” overall use these methods, with “Mask R-CNN” going even deeper by enabling pixel-level instance segmentation, allowing greater precision (Han, 2025). Although this method offers state-of-the-art accuracy, the two-step process introduces complexity that significantly increases computational resource usage, thereby bottling up real-time detection and limiting the range of hardware it can run on.

The main difference among these architectures is the Region Proposal Network (RPN). Unlike single-stage models that predict bounding boxes across the entire image at once, the RPN serves as an initial filter. It moves a small window across the image feature map to create "anchor boxes" for possible objects. Only regions with high "objectness" scores go on to the second stage for further classification and refinement. This process improves accuracy but also highly increases

computational delay, as the network must perform two separate, resource-intensive forward passes per image (Ren et al., 2015).

1.2.3. Unified (Single-Stage) models

Unified models approach the problem in a completely different philosophy. What if the network “glanced” rather than “scanned”? This primary difference in paradigm allowed to change the problem back into a single regression problem. Popular Architectures like the YOLO series process the entire image in a single forward pass, predicting the area and box of the object of interest, and give a statistical probability or confidence of this mentioned object. This architecture has been critical for numerous autonomous systems that rely on low-latency detection and classification. In the context of robotics, this enables the system to process visual data and react to dynamic environments (Manikanta & Mehershilpa, n.d.). Navigating around obstacles or tracking defects in real-time, without the latency penalties associated with multi-stage detectors, enables these systems to work seamlessly.

Comparison between YOLOv8 and Mask-R-CNN was done on “Fractured Endodontic Instruments,” and the results clearly show that YOLOv8, which is already an outdated model for today’s standards, not only outperformed in accuracy with near-perfect results but also did so in a fraction of the time that it took Mask-R-CNN to do so. These results are shown in Table 1.1 (Çetinkaya et al., 2025).

Table 1.1. Comparison of performance metrics between YOLO v8 and Mask-R-CNN models for detection and segmentation (Çetinkaya et al., 2025)

	Accuracy	IoU (Intersection Over Union)	mAP50	Interference Time
YOLO v8	0.973975	1.00	0.989	14.6 ms
Mask-R-CNN	0.982075	0.99	0.950	88.7 ms

For solar panel inspection, this trade-off creates a clear divide in application. Two-stage detectors like Mask R-CNN are often used for "offline" processing, where a drone lands and uploads terabytes of data to a server. Powerful GPUs then spend hours analysing 4K images to find hairline micro-cracks. In contrast, these detectors are rarely used on the drone itself because they cannot process video feeds in real-time. Recent 2025 benchmarks, shown in Table 1.1 above, highlight this gap. While Mask R-CNN offers competitive accuracy, its inference time is nearly six times slower than modern single-stage alternatives like YOLOv8.

1.3. Natural language processing and Agentic AI

Natural Language Processing (NLP) provides the historical foundation for converting unstructured language into representations that machines can process. Through pipelines such as tokenization, parsing, and information extraction, NLP shifted from hand-crafted rules to learned representations, such as vector embeddings and transformer-based models. With these improvements, the field converged into Large Language Models (LLMs) that can perform a broad scope of tasks within a single general-purpose architecture. Within the scope of this thesis, LLMs are important not only because they are expected to detect and classify defects, but also to express findings in a human-readable form and create actionable plans to solve the issue at hand. Agentic AI builds on LLMs' capabilities and improves them by creating a controlled environment in which multiple agents (or LLMs) can work on the problem and systematically solve it to ensure the quality of the output. This systematic, goal-driven approach, combined with tools for these LLMs to broaden their scope and capabilities, makes the user-facing output significantly more valuable than just defect detection (Narinder Pal Verma, 2025).

1.3.1. Large language models

With the AI race underway, extremely capable and state-of-the-art models coming out almost every month, and the sheer number of options we already have, it can be challenging to pick a model for our system. There are multiple parameters to consider, such as costs, limits, and the context window. Thanks to the fierce competition between OpenAI, Google, Anthropic, and many more. These models get bigger, more efficient, and more intelligent every day. For our systems use case, the main parameters we are interested in are its cost, context window, and ability to accept images as a context for the analysis.

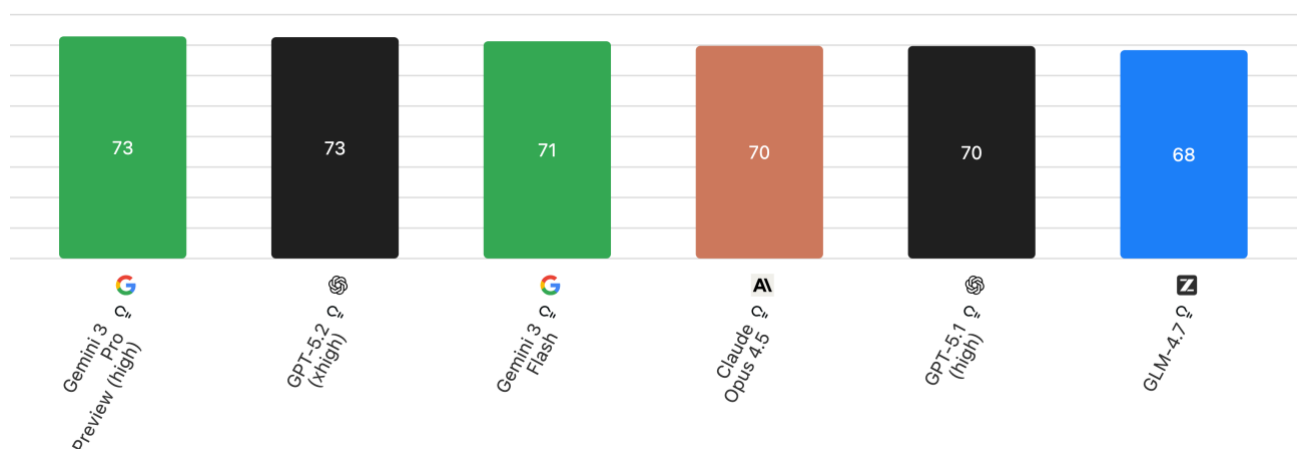


Fig. 1.10. Artificial Analysis Intelligence Index of frontier models (Artificial Analysis, n.d.)

As you can observe in Fig. 1.10 (*Comparison of AI Models across Intelligence, Performance, Price* | *Artificial Analysis*, n.d.), there are numerous companies and evaluation methods to rank LLMs and compare their performances, but according to the Artificial Analysis Intelligence Index, the highest-ranking models as of 4th of January 2026 are Google's Gemini 3 Pro (high) and OpenAI's GPT-5.2 (xhigh). This index is the cumulative score across 10 independent evaluation methods. These methods are: MMLU-Pro, GPQA Diamond, Humanity's Last Exam, LiveCodeBench, SciCode, AIME 2025, IFBench, AA-LCR, Terminal-Bench Hard, and τ^2 -Bench Telecom. They measure these models' capabilities by querying them extensively across domains, including general world knowledge, physics, law, history, biology, chemistry, scientific questions, logical questions, interpretations, real-world capabilities, etc. There are even queries that are known not to be answered correctly, but they measure the model's thought process, hallucinations, and the way it arrives at the answer. They are standardized and given to every model under the same circumstances to keep the objectivity and the credibility of the evaluations (Bragg et al., n.d.; *Humanity's Last Exam Benchmark* - by Nilesch Barla, n.d.; Jain et al., 2025; Pyatkin et al., 2025; Tian et al., 2024; Wang et al., 2024). It is also worth noting that some LLMs are more domain-specific. Therefore, they might be a better performing model for a niche use case, but perform poorly and not make it to the top ranks when the cumulative scores are involved. These models are advantageous across domains, mostly due to their large size and parameter count. The majority of the top performers are proprietary models, but they are suspected to have billions or even trillions of parameters. As a point of reference, the highest-performing open-source model among them, GLM-4.7, has around 358 billion parameters. This admittedly does not give us an exact number of what other models are doing, but it is enough for us to picture their parameter count as around that number or larger.

The combination of LLMs into workflows completely changes the approach for raw data reporting. Instead of reporting the direct data output of a problem, reporting the explanation, solution, and step-by-step guide for performing it completely autonomously revolutionizes maintenance systems. Since this solution is intended for residential scenarios, guiding the owner would improve the user experience immensely. For instance, operators might get overwhelmed or not exactly know the cause of the problem or solution they are facing; these systems could reduce the cognitive effort required of operators and focus on the actual solving rather than trying to interpret the problem itself.

1.3.2. Agentic frameworks

While LLMs provide the reasoning capability, Agentic Frameworks provide the execution structure. A standard automation script follows a rigid, linear path. If an unexpected obstacle appears, the script fails. In contrast, an Agentic system employs the "ReAct" (Reason + Act) method, which allows the system to dynamically decompose high-level goals into logical sub-tasks and approach

them systematically by querying itself and having a thought chain, as seen in Fig. 1.11 (*What Is a ReAct Agent? | IBM, n.d.*).

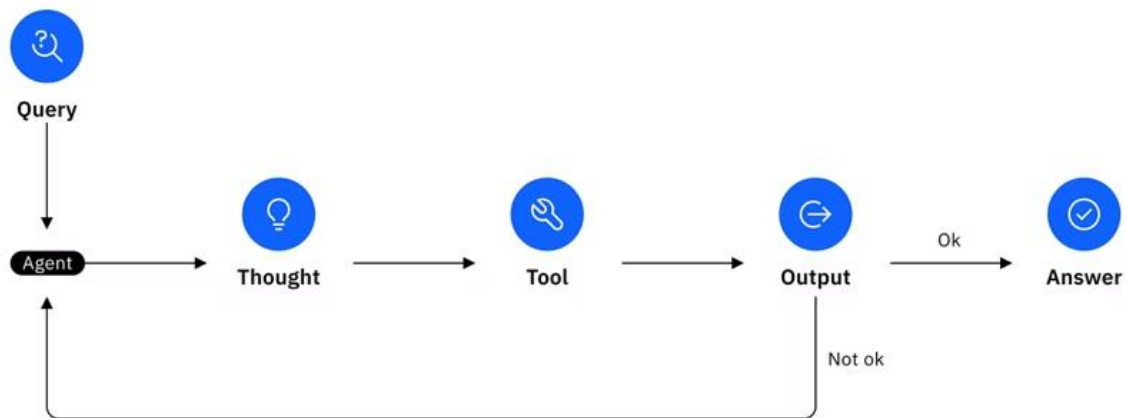


Fig. 1.11. React visual representation (*What Is a ReAct Agent? | IBM, n.d.*)

“ReAct” pattern works significantly different than a single LLM call. It basically is an iterative loop that positively reinforces itself by the self-generated content of the agent. The loop consists of three main steps. Thought -> Action -> Observation. The initial step, “thought”, analyzes the users prompt and creates an action plan to iterate on later. These are mostly actionable steps that the agent separates to keep track of the progress and make the overall task-solving problem systematic. After dividing the main task into approachable subtasks, the second step comes in the main loop. “Action” is where the LLM gets things done via tool calls. Tools are executable scripts in a programming language that perform deterministic actions more reliably for agents. Agents utilize these scripts in their own runtime in Agentic frameworks for various reasons. This could be from getting more data or running actions in an external system via communication protocols such as MCP (Model context protocol), A2A (Agent to Agent), or just a simple local script to a weather API. After the agent executes an action, it needs to ensure that everything is on track. That is the “Observation” which controls the end of an iteration and determines whether that step was done correctly and what the next step is. If the agent determines that a particular step was successful, we move on to the next item on the list. It is worth noting that the observation step could be done more reliably if we can ensure that an action in an external system was performed via another tool. Although this is not mandatory, it is an industry best practice. This whole explanation can be summarized as an internal prompt for an agent. Here is a sample prompt displayed in Fig.1.12 (*What Is a ReAct Agent? | IBM, n.d.*).

```

Answer the following questions as best you can. You have access to the following tools:

Wikipedia: A wrapper around Wikipedia. Useful for when you need to answer general questions
about people, places, companies, facts, historical events, or other subjects. Input should be
a search query.
duckduckgo_search: A wrapper around DuckDuckGo Search. Useful for when you need to answer questions
about current events. Input should be a search query.
Calculator: Useful for when you need to answer questions about math.

Use the following format:

Question: the input question you must answer
Thought: you should always think about what to do
Action: the action to take, should be one of [Wikipedia, duckduckgo_search, Calculator]
Action Input: the input to the action
Observation: the result of the action
... (this Thought/Action/Action Input/Observation can repeat N times)
Thought: I now know the final answer
Final Answer: the final answer to the original input question

Begin!

Question: {input}
Thought:{agent_scratchpad}

```

Fig.1.12. Sample prompt for ReAct framework (*What Is a ReAct Agent? | IBM, n.d.*)

Moreover, the “ReAct” framework depends on a previous prompting technique called “Chain-of-thought prompting”. This is a proven method for getting more reliable results from LLMs. The essential idea of a systematic approach remains the same; ReAct builds on it with improvements to observation and validation for more production-grade solutions (Wei et al., 2022).

Like prompting frameworks, there are many popular agentic frameworks, but for our use cases, the most relevant ones are “LangChain” (LangChain, n.d.), “LangGraph”, “CrewAI” (CrewAI, n.d.), and “LlamaIndex” (LlamaIndex OSS, n.d.). It is worth mentioning the incredible communities behind these frameworks that have created these production-grade frameworks and kept them open source. FOSS (Free and Open-Source Software) has enabled continuous improvements and prevented a single framework from monopolizing the industry. The LangChain framework is mostly known for its low-level control structure, and its LangGraph extension allows developers to define agents as nodes so that when a node fails, it can fail gracefully, providing fine-grained control over the system. This makes the system more reliable and predictable in unexpected situations, which makes it a great framework to use in production use cases. CrewAI, on the other hand, is built on top of LangChain. It abstracts the complexity of LangChain’s low-level approach into simpler concepts like tasks, agents, and crews. An agent’s role is defined by the attributes assigned to it, including role, goal, and backstory. This approach helps the agent tackle the given task with this context. Therefore, multiple

agents in a crew with this setup work in a way that is very much like a real-world team scenario, where there is a manager, the orchestrator in our crew case, that oversees the workers (or other agents). Also worth mentioning that CrewAI has “ReAct” approach baked in to its system, abstracting the need for creating a custom setup for each project.

Table 1.2. Comparison summary of different Agentic AI framework (Roberto Nava | Medium, n.d.)

Criterion	LangChain	LlamaIndex	CrewAI
Core Philosophy	Flexible Orchestration: A "framework for everything" that connects AI components.	Data-First (RAG): The most effective way to bring your data to LLMs.	Agentic Collaboration: Multiple agents working together autonomously.
Ideal Use Case	Rapid prototyping, complex chains with multiple tools (APIs, calculators), general-purpose applications.	Chatbots over documents, Q&A systems, knowledge bases	Automating complex workflows (e.g., market analysis), multi-step content generation, team simulations.
Learning Curve	Moderate to High. It's easy to start, but mastering its abstractions (LCEL, "LangChain spaghetti") takes time.	Low to Moderate. For standard RAG use cases, it's incredibly direct and optimized.	Low to Moderate. The "Roles" and "Tasks" abstraction is very intuitive, simplifying multi-agent system creation.
Flexibility	Very High. It's the "Lego" of LLM frameworks. You can build almost anything, but you have to connect the pieces.	Focused. Extremely powerful in its niche (RAG), but less of a generalist than LangChain for pure orchestration.	Structured. Guides you toward a specific design pattern (collaboration), which makes building easier but is less flexible.
Abstraction	Low-level components (LLMs, Prompts, Parsers) and high-level ones (Chains, Agents).	High-level abstractions centered on the data pipeline (VectorStoreIndex, QueryEngine).	Very high-level abstractions (Agent, Task, Crew) that encapsulate the underlying complexity.

Although these frameworks are flexible for almost any AI use case, there are more opinionated frameworks for more specific use cases, like LlamaIndex. It focuses primarily on data agents rather

than workflow orchestration. It excels at connecting your data to LLMs for knowledge retrieval tasks, such as querying a PDF manual you have, instead of trying to find your specific problem after scanning the whole page manually. Finally, for a more concise summary, please refer to Table 1.2 below (LangChain vs LlamaIndex vs CrewAI by Roberto Nava | Medium, n.d.).

1.4. Substantiation of the proposed solution

As can be seen from the present condition of the solar panels maintenance market, there is a certain “scale bias” when new technologies are predominantly developed for large-scale solar stations while residential and small commercial areas are almost entirely ignored (Akrofi et al., 2024). This paper aims to suggest a cost-effective solution based on cheap hardware and agentic AI. The following part of this paper will focus on the market need for the technology suggested, its feasibility and economic sustainability.

1.4.1. The residential market gap

Cost rather than technology is the core issue for effectively maintaining residential solar energy systems. Because conventional methods such as manual inspection and thermographic drone surveys are only economical when used on a large scale, these types of service generally cost more than the actual efficiency gains they provide a typical person or household with a 5kW system. This leads many homeowners to take a waiting approach until they experience some type of system failure. However, various environmental factors, such as dirt, shading and hotspots can drastically reduce an entire year’s worth of energy produced from a solar system. For example, studies conducted by NREL and other organizations have shown that just from dirt washing off of their solar, it will reduce potential yearly energy production by 3% to 7%. On top of this, many local issues such as bird waste causing partial shading of the panels can result in around 20%-50% of power being lost at the string level due to mismatched electrical characteristics. (Bovarnick & Johnson, n.d.; Van Opstal & Smeets, 2023). This demonstrates that there would be a large demand for a permanent low maintenance solution to provide ongoing automated monitoring solutions for these systems. By developing a hardware system that could cost the same as one professional inspection, it would create an opportunity to develop a proactive rather than reactive approach to maintenance and provide solutions to eliminate long-term problems that shorten the useful lifetime of their solar electric systems, improve their efficiency, and increase their energy output per unit produced. Adequate coverage and hardware feasibility.

Choosing the Raspberry Pi 5, and pairing it with the Camera Module 3 Wide has been shown to perform well enough for real-time edge inference. Benchmark tests show the Raspberry Pi 5 can run quantized YOLOv11n models around 10-25 frames per second with the help of optimization tools

such as NCNN or OpenVINO. As a result, it would be possible to perform effective real time defect detection on readily available and inexpensive hardware, rather than requiring expensive accelerators such as NVIDIA Jetson. Additional notable features of the Raspberry Pi Camera Module 3 include having a 12-megapixel (12MP) Sony IMX708 sensor with an autofocus option (Raspberry Pi Camera Module 3 Standard NoIR Wide NoIR Wide, 2024a), an example of which can be found below in Fig. 1.13. The optical quality of the camera is suitable for the identification of surface defects. With a single wide-angle camera mounted on a tripod between 1.5 and 2 meters away from a target, it is possible to view either a 2x2 panel array or a 3x2 panel array.



Fig. 1.13. Raspberry Pi Camera Module 3 (Raspberry Pi Camera Module 3, 2024)

This arrangement allows the system to digitally zoom into specific regions of interest (ROIs) to detect macro-defects such as heavy soiling, bird droppings, or physical damage. It addresses the resolution gap between close-range handheld inspections and high-altitude drone surveys.

1.4.2. Detection and logic

To address the competing needs of data privacy, latency, and computational intelligence, the proposed approach uses a "Scan and Sleep" method. The "Scan" part is located at the edge, where a Raspberry Pi runs a lightweight YOLOv11n model with a snapshot interval adjustable by the user, defaulting to 5 minutes. Alternatively, users can choose a continuous live feed where the local model handles detection, keeping neutral video feeds private and reducing bandwidth. When detection surpasses a set threshold, a snapshot is sent to a public cloud provider via LLM calls within our agentic framework for additional analysis.

1.4.3. Agentic workflow with CrewAI

Whereas rigid automation scripts are executed on a straightforward "if-this-then-that" basis, this system relies on an Agentic-AI workflow based on the CrewAI framework and the Gemini 3 Pro model. This acts just like a human work team by allotting various roles. A Defect Analyst Agent looks at the computer vision results first, using its Visual Language Model (VLM) capabilities to

interpret findings and nullify false positives, such as shadows or passing animals. Then, a Maintenance Planner Agent checks actual defects against certain external factors; for example, if it indicates "heavy dust," while then considering whether rain is imminent based on weather data, it would recommend delaying cleaning. Finally, a Report Writer Agent compiles these findings into an email for the homeowner that is quick and easy to read about the severity of the issue, up to the potential efficiency loss, and provides the attached photo. This whole process of multiple agents is, in effect, an ensemble transformation of raw detection data into a well-rounded, useful insight for field operations.

1.4.4. Economic viability and ROI

This system has a significant economic advantage compared with standard operating costs associated with maintenance. The fully assembled device (i.e., Bill of Materials) with a Raspberry Pi 5, 3D camera, AM2302 sensor, and custom housing amounts to about 210 EUR as an identifiable capital project expense and represents a relatively comparable amount of money compared with one professional maintenance visit. Ongoing operational costs are very affordable due to low electricity costs and lower API usage fees for the Gemini Flash model. Preventing long-term effects such as potential induced deterioration (PID) and hot spots will extend the life of the asset in question and ultimately improve daily energy production. Therefore, the projected return on investment (ROI) for this system should occur within 1 to 2 years, presenting a significant financial advantage to an entity by not maintaining systems or relying solely upon costly manual repair methods.

2. CALCULATION OF THE PROJECT

The distributed architecture and the selection of compatible components are vital to bringing a system like this to life. Given our computational, environmental, and cost constraints, our decisions must be deliberate. The choice of the YOLOv26n computer vision model and the inclusion of a computer vision component in the project stem from the smart triggering system implemented to make LLM calls as efficient as possible. Therefore, the model serves as a periodic filter, ensuring the system uses its full compute only, when necessary, while the agentic workflow is active. The LLM component of this system serves as the primary interpreter for the complex agentic workflow demonstrated using 4 agents. The freedom in model choice is an architectural decision that allows the end user to bring their own keys, though initially it is assumed that the Gemini free tier is being utilized, given its generous free tier offer and the models' capabilities across dimensions with only one API (Application Programming Interface) key.

2.1. Hybrid edge and cloud architecture

In designing this system, we paid the utmost attention to self-reliance and cost efficiency. Therefore, this system almost runs everything locally or within the free range offered by popular service providers. The lack of monthly fees for the device reinforces its claim to be cost-efficient and an overall easier to set-up system. Starting with the local computer vision model running YOLO 26 nano, built specifically for low-power edge devices, to detect issues on the solar panels and act as a trigger for our agentic workflow. Then, the CrewAI crew runs locally on the Raspberry Pi as an orchestrator but sends LLM requests to Google's Gemini API. This hybrid approach enables us to eliminate monthly hosting fees for our agent runtime and the computer vision model, while also keeping us within the free tier of the Gemini API thanks to our smart trigger system. The crew's comprehension occurs only when the computer vision model flags something, and it runs only periodically. This period can be adjusted by the user via the “Solar Sentinel” management dashboard. This dashboard is also hosted locally on each device, ensuring the system is distributed and as secure as possible. There will also be an option to access it from any network via “Tailscale”. The self-hosted dashboard “Solar Sentinel” will be an all-in-one control and management dashboard for your device.

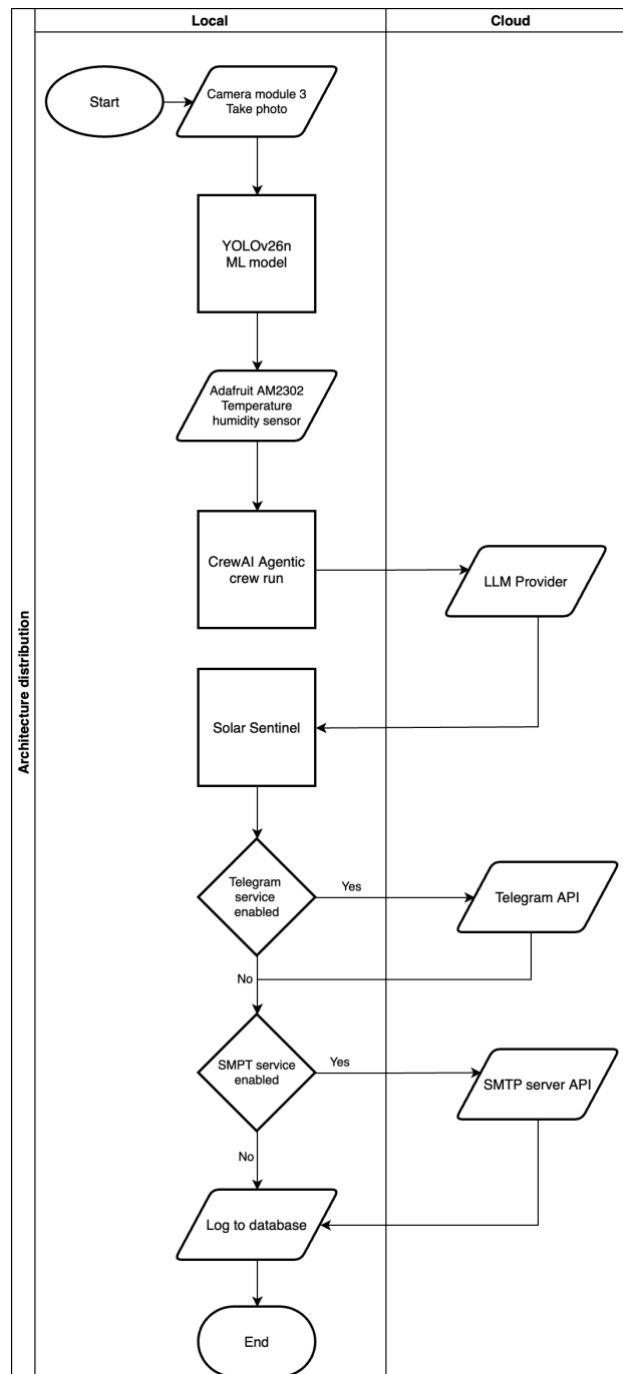


Fig. 2.1. Architecture distribution diagram.

Furthermore, the architecture distribution is better illustrated in the diagram in Fig. 2.1. All the services are locally hosted and provided, except the LLM provider and the notification servers, such as Telegram API (*Telegram APIs*, n.d.) for messages or SMTP (SMTP Server Explained - AWS, n.d.) server for emails. Additionally, “Tailscale” (Tailscale, n.d.) could be added to the diagram, but since it’s a completely optional way of interacting with Solar Sentinel, it’s not significant enough to be included in the general architecture distribution diagram.

2.2. IMPLEMENTATION OF AI MODELS

The term Artificial Intelligence (AI), as broad as it is, has fundamental importance in modern society. Within the scope of this system, there are 2 different aspects under the umbrella of AI. The computer vision model and the agentic workflow. Each is more important than the other, making comprehension and accurate reporting possible to the end user. Although the majority of the heavy lifting is done by the agentic workflow, thanks to the computer vision model, the system remains cost-effective enough to stay within the free-tier limits of LLM providers.

YOLOv26n, released in early 2026, is the state-of-the-art version of the infamous YOLO family. The “n” in the appendix indicates that it’s a “nano” model, which is optimized for edge hardware with power constraints. It features a native end-to-end prediction that bypasses the need for Non-Maximum Suppression (NMS) post-processing, making this variant 43% more efficient as shown in the Fig. 2.2. Due to the separation of representation learning from heuristic post-processing and using the Momentum-unified stochastic gradient descent (MuSGD) optimizer for training stability, the accuracy of the model is high even for small-target detection while running efficiently on CPU-only edge devices(Chakrabarty, 2026). The model is used only as a trigger for the system, not as the inference layer. Therefore, it performs only binary classification between “healthy” and “defect” to reduce its computational burden. This decision was made mostly because no matter what kind of detection type has triggered the system, the same agents run as an inference layer that further comprehends the problem, and that’s where the “intelligence” of this system comes from. The computer vision part is merely a smart trigger system.

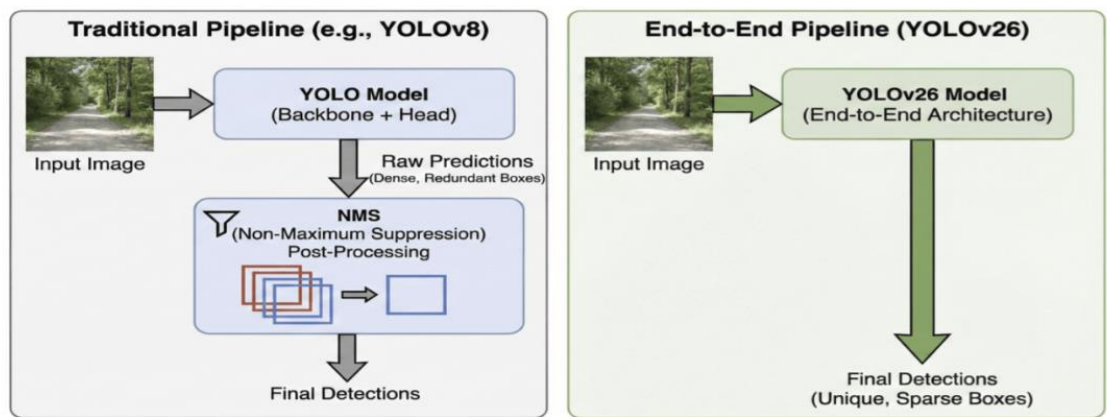


Fig. 2.2. Architectural differences between YOLOv8 and YOLOv26(Chakrabarty, 2026)

The impact of this architectural difference can also be observed in between previous models and YOLOv26 in terms of latency. This difference in between the same model family and other contenders is clearly highlighted in Fig. 2.3.

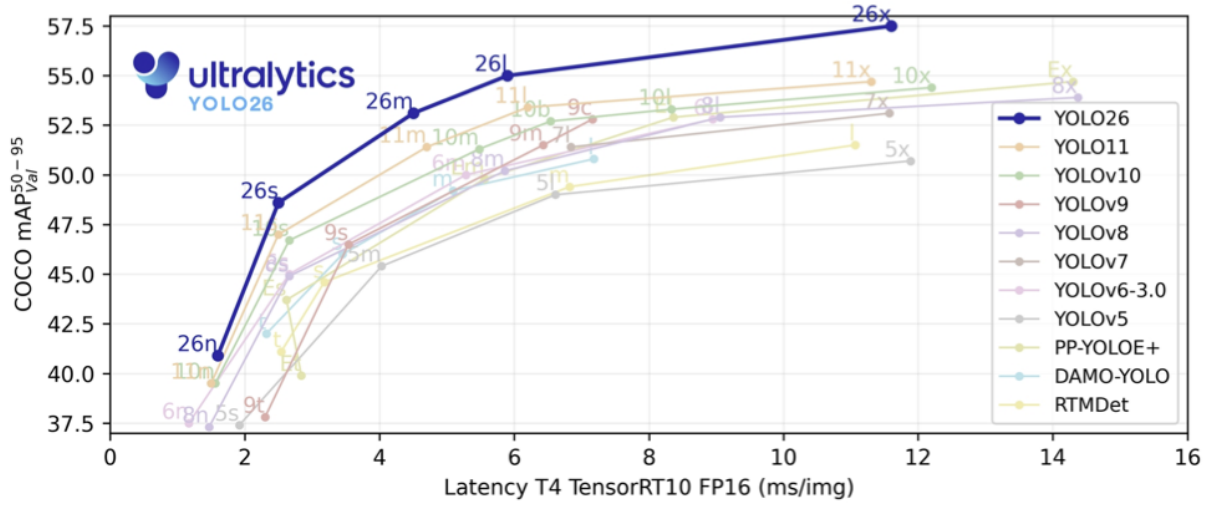


Fig. 2.3. Latency of computer vision models in COCO datasets with T4 TensorRT10 (Chakrabarty, 2026)

Although the model is designed for edge computing and operates with low latency, the system has severe computational constraints. The model should run without triggering the Raspberry Pi's throttling limits. To do that, it undergoes 8-bit integer quantization, which significantly reduces the memory footprint in the LPDDR4X RAM of our microcomputer. To fully understand why this model can run on edge devices despite thermal and computational limits, it is necessary to examine the model's structure in detail. The YOLOv26 detailed architecture (Hidayatullah & Tubagus, 2026), as shown in Fig. 2.4, divides its processing pipeline into 3 important stages. The initial stage is the "Backbone," which serves as the foundational feature extractor. It performs initial processing of the raw input image by iterating through a sequence of standard convolutional layers and specialized C3k2 blocks. These blocks reduce the spatial dimensions of the image while identifying meaningful, complex geometric patterns. These properties help the model identify defects, such as cracks or dust patterns, on photovoltaic cells without requiring large amounts of parallel processing power.

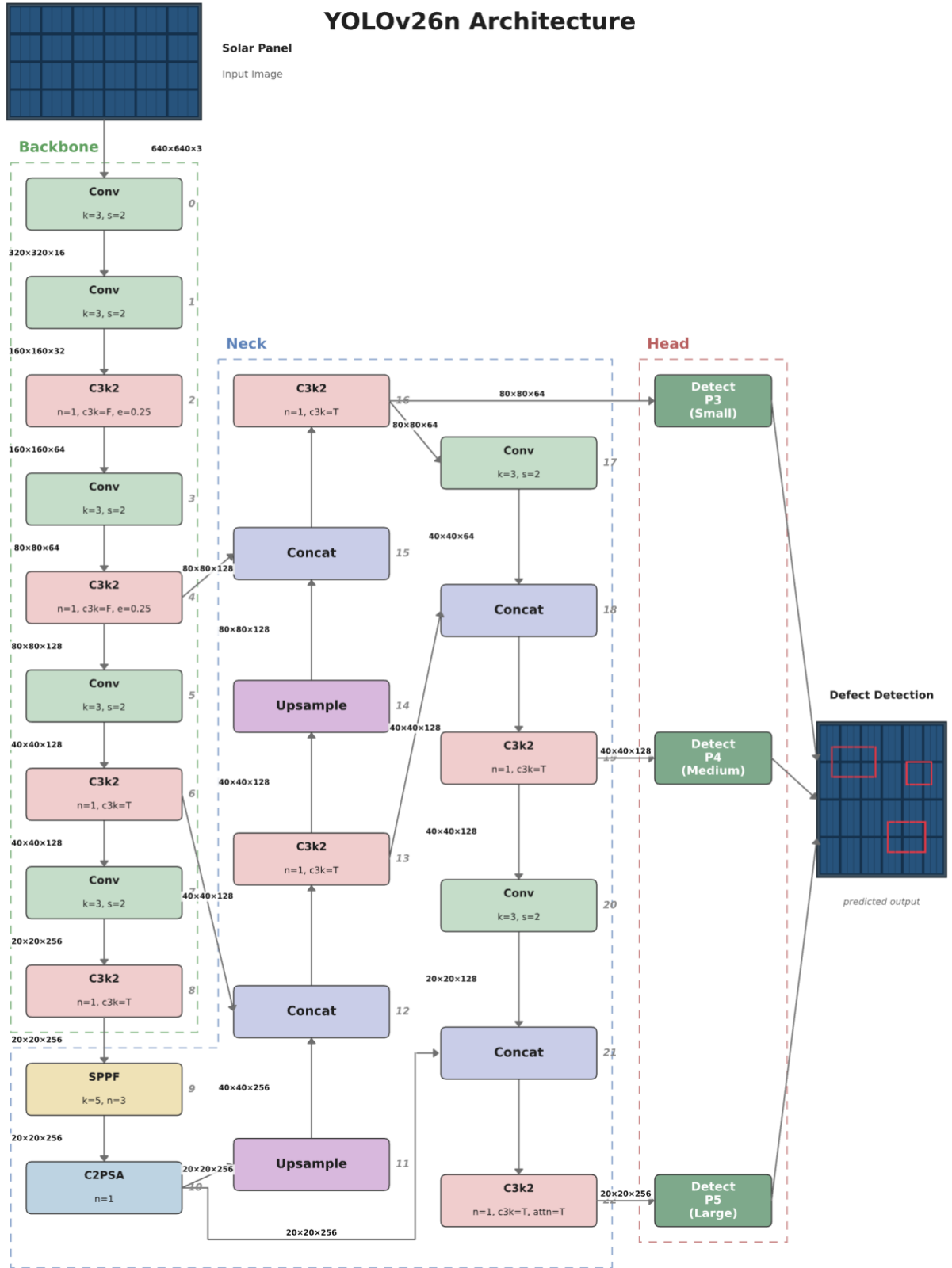


Fig. 2.4. YOLOv26 detailed architecture diagram

Secondly, after the initial features are extracted, they are moved to the next layer which is called the “Neck”. This middle stage acts as a feature aggregator. It heavily relies on the SPPF (Spatial

Pyramid Pooling Fast) module, having been updated to have a faster shortcut connection, this mechanism is crucial for low power devices. Allowing abstracted semantic information to bypass certain processing bottlenecks, it stabilizes the gradient flow and prevents data loss during the aggressive down sampling phases. Throughout SPPF, a number of up sampling and combination operations merge features from different abstraction levels. This ensures the model keeps the fine-grained information especially important with detailed tiny surfaces.

Finally, the data reaches the “Head” stage of the processing pipeline. This stage is fully responsible for generating the final predictions after all the abstractions, upsampling, and downsampling. An important characteristic of the YOLOv26 architecture is its aggressive optimization within the detection blocks. Previous generations favored a method called DFL (Distributed Focal Loss) to calculate the probabilities of bounding boxes. This method has significant computational overhead. Instead, the current architecture eliminates this entirely and favors direct box regression. It is a streamlined approach that calculates the coordinates of an object without the need for complex probability distributions (Hidayatullah & Tubagus, 2026). Therefore, this architectural shift significantly reduces latency, allowing our system to rapidly classify the image as “defective” or, if nothing is detected, classify it as “healthy”.

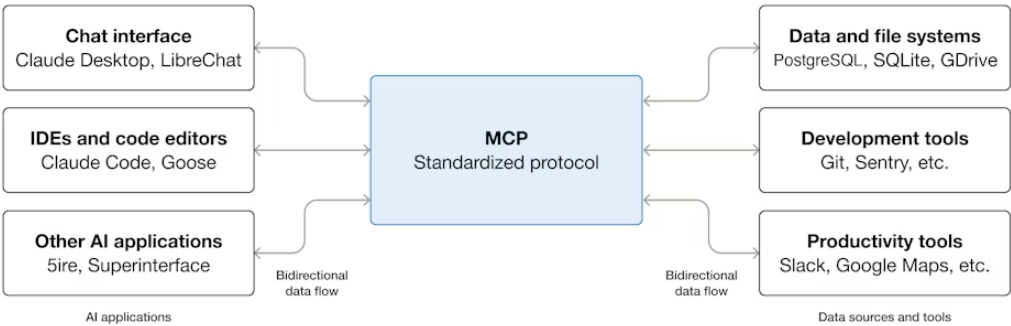


Fig. 2.5. MCP feature diagram (Model Context Protocol, n.d.)

Moreover, the system uses the MCP protocol created by Anthropic (Model Context Protocol, n.d.) as a standard way to connect external systems to our agentic orchestration. There are many reasons to use MCP, such as security. MCP enforces a strict client-server architecture that keeps tools like Python scripts and the agent runtime separate, protecting the runtime against prompt injection attacks. The main advantage is that it provides a standardized layer compatible with existing solutions, as seen in Fig. 2.5, allowing connection with just a few lines of code.

2.3. Mechanical design and thermal management

Running a computer vision model and an agentic orchestration framework locally on a Raspberry Pi generates a substantial thermal load. To address this thermodynamic challenge, the enclosure is manufactured using ASA (Acrylonitrile Styrene Acrylate). This material was selected for its high glass transition temperature of 105 degrees Celsius (Akıncioğlu & Aslan, 2022), ensuring the chassis resists deformation under heat or sunlight. Furthermore, the design incorporates passive ventilation slots to facilitate natural convection and efficient air exchange. As illustrated in Fig. 2.6, these horizontal exhaust apertures measure 30 mm by 4 mm on the rear panel and 80 mm by 4 mm on the bottom panel. The enclosure's overall footprint of 160mm in length, 112mm in width, and 52mm in height provides a large internal air volume, ensuring an active cooling mechanism, such as a fan, can operate efficiently.

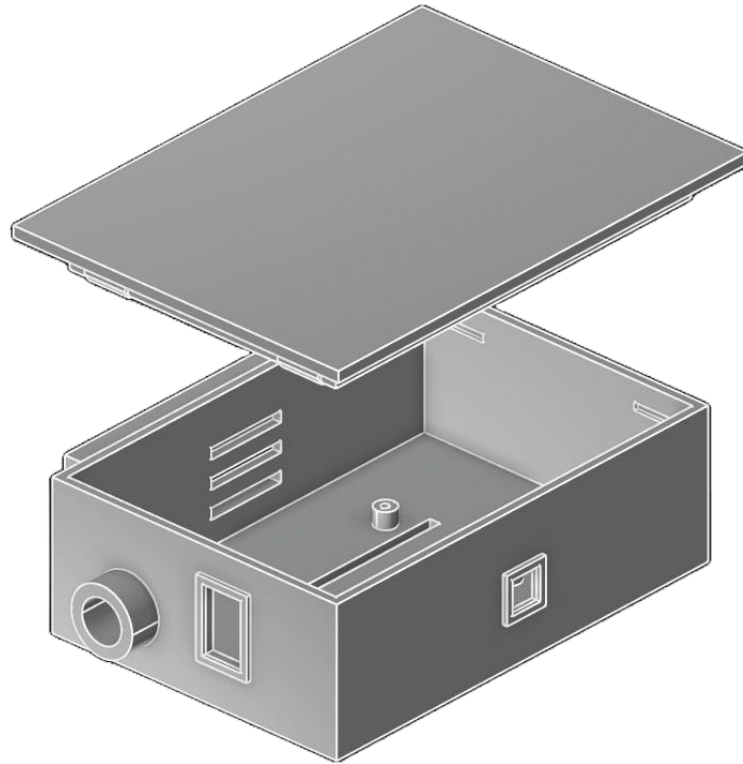


Fig. 2.6. Exploded view of the custom enclosure

Moreover, a highly robust wall thickness of 3.2mm provides a strong structural integrity and rigidity when printed with ASA. The 8mm lid overhang ensures that liquids like rainwater won't drip directly off the enclosure roof, and the gasket groove seals via a snapping arm mechanism that pulls the lid tight, providing the necessary sealing pressure to protect the electronics inside. It is worth mentioning that both thermo/humidity sensor mount and camera mount are tightly fitted for stability

and preventing foreign substances from getting inside the casing. The Raspberry Pi 5 module is screwed via Pan Head (ISO 7045) screws that are sized M2.5 to the enclosure and the active cooler is then mounted to the top of the board using its integrated spring-loaded screws. Lastly, the 15.5 mm diameter power cable gland extending out of the enclosure reduces the chance of water droplets entering the system.

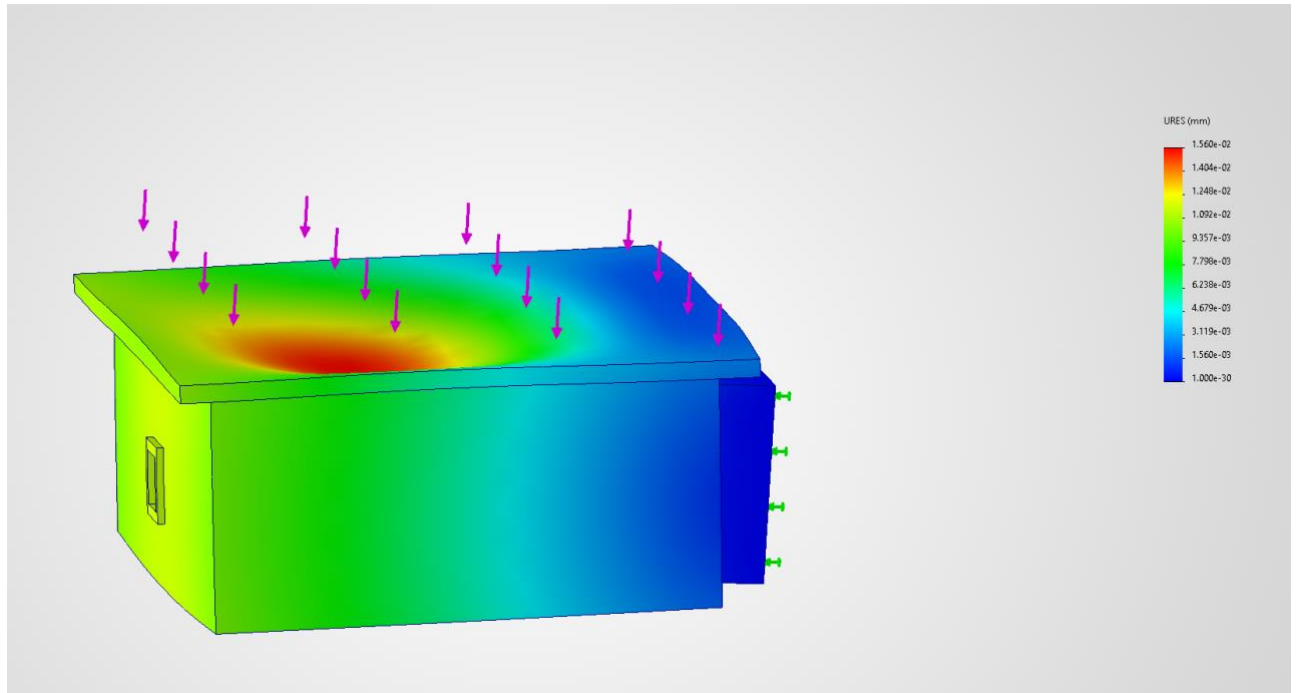


Fig. 2.7. Resultant displacement (URES) results of custom enclosure

Distribution of the enclosure under a 10-newton distributed vertical load was analyzed using Finite Element Analysis (FEA) in SolidWorks Simulation. This simulation models a realistic scenario where the enclosure is mounted from the rear mount area, including the fixture. The force is directed from top to bottom onto the lid. The color contour map shows deformation magnitude across the assembly, with red indicating maximum displacement and blue indicating minimal movement. The highest displacement, 0.0156 mm, occurs at the unsupported corner of the lid, farthest from the fixed mounting boundary. This aligns with behavior expected of a plate fixed along one edge and free along the others, where the free corner deflects most under uniform load. The small maximum displacement relative to the enclosure's size suggests it retains its shape under load, avoiding significant risk of mechanical failure under normal operation.

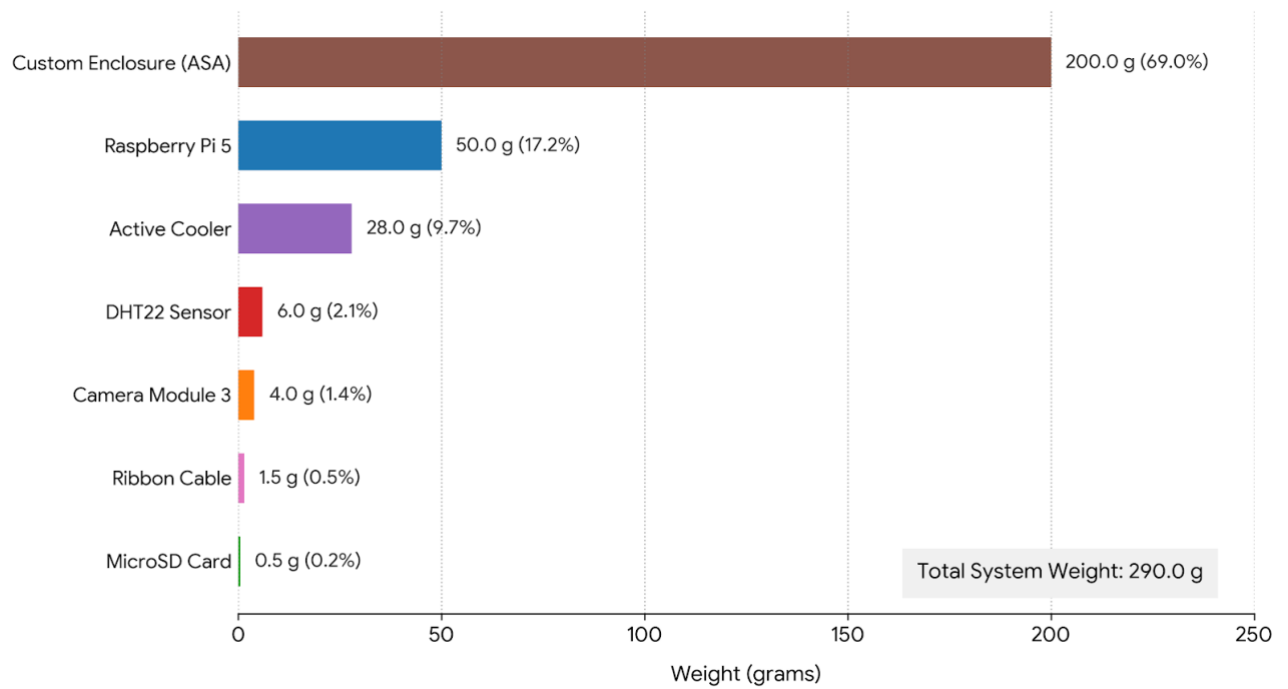


Fig. 2.8. Assembly components weight distribution horizontal bar chart

Overall, the entire assembly weighs around 290 grams, and the weight distribution is shown in Fig. 2.8. The custom enclosure is the heaviest component in this assembly, including the “Case” and the “Lid,” both custom 3D-printed in ASA, totaling around 200 grams. Next comes the microcomputer, the central computing system on which everything runs, adding 50 grams. Then the active cooler, consisting of two parts, the active fan and the passive heat dissipator, adds 28 grams more to the total. Due to the heavy reliance on edge computation in the system, an efficient cooling mechanism is utmost important. Moreover, the sensors, such as the Adafruit DHT22/AM2302 thermal and humidity sensor and the Raspberry Pi camera module 3 wide, follow with 6 grams and 4 grams, respectively. Finally, other miscellaneous parts, such as the ribbon cable for the camera module and the MicroSD card, weigh the least. It is also worth mentioning that other potential weights, such as the various number of screws in the process of assembling, can add negligibly to the system’s total weight.

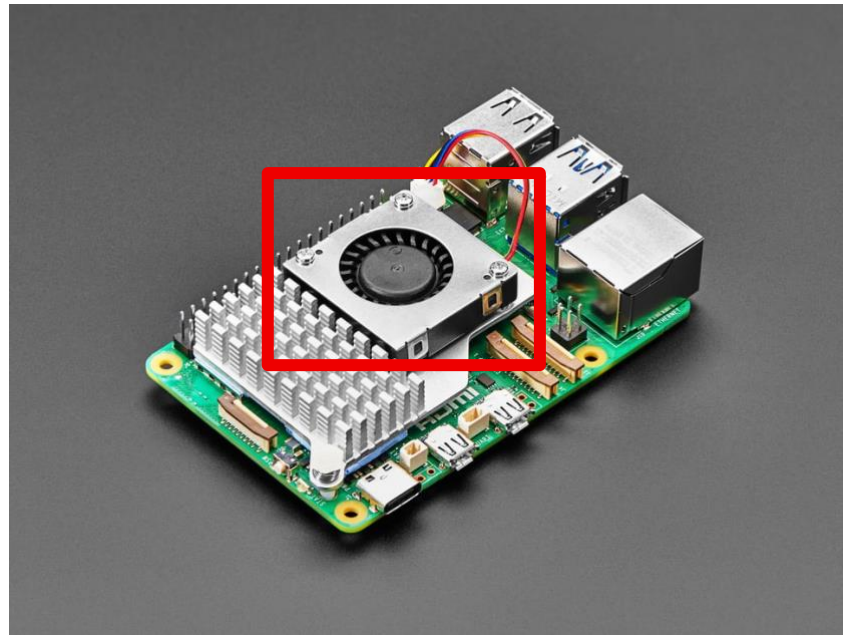


Fig. 2.8. Raspberry Pi 5 active cooler mounted (Adafruit Industries, n.d.)

Passive cooling methods via vents are effective but not as efficient as an active cooler like the Raspberry Pi 5 active cooler (*Raspberry Pi Active Cooler for Raspberry Pi 5*, 2026). It is a dedicated, permanent mount solution designed specifically for the Raspberry Pi 5, requiring 5V DC supplied via the Raspberry Pi 5's four-pin fan header. The active cooler combines an aluminium heatsink with a temperature controlled blower fan to keep the system at comfortable operating temperatures even under heavy loads. It stabilizes on top of the Raspberry Pi 5 via spring-loaded push pins and comes with pre-applied thermal pads for efficient heat transfer. The cooler also features volumetric airflow of 1.09 CFM to pump as much heat away from the microcomputer as possible, a maximum speed of $8000 \pm 15\%$, and a long mean time to failure of 379000 hours at 25 degrees Celsius. With these features in mind, the decision to include the active cooler is not only necessary for a faster system but also increases its overall lifetime, since it runs at more comfortable temperatures.

3. DESCRIPTION OF THE CONSTRUCTION AND OPERATIONAL PRINCIPLE

To have a maximally capable development environment with a small form factor and a reasonable price, we have chosen the Raspberry Pi 5 8GB as our main board. Moreover, to avoid potential compatibility issues, we have chosen Camera Module 3 Wide. Running our computer vision model locally alongside an agentic workflow required us to choose at least the 8 GB version of our main board. To run everything simultaneously at reasonable temperatures, we have decided to use the Raspberry Pi Active Cooler for the Raspberry Pi 5. Lastly, to be more self-reliant, we have chosen the Adafruit AM2302 temperature and humidity sensor to measure our own temperature and humidity. Analyzing the system piece by piece doesn't reveal the full picture until all parts come together to animate the system. Operational logic is as important as the technologies used within a system. From system triggers to cost-saving techniques in the main app logic, the backend is essential for unifying this logic. From data operations to integrations between different parts, "FastAPI" serves as the control hub of this system. It manages the sequential execution of the entire workflow with error-handling mechanisms and robust endpoints.

3.1. Hardware selection and capabilities

Not relying on cloud computing requires a powerful edge device. The Raspberry Pi 5 (*Raspberry Pi 5*, 2026) is the flagship model in its series. It features a Broadcom BCM2712 quad-core ARM Cortex-A76 processor running at 2.4 GHz. Its processing power is three times that of the previous-generation Raspberry Pi 4B (*Raspberry Pi 4 Model B*, 2026), which uses a Broadcom BCM2711 quad-core Cortex-A72 processor at 1.8 GHz. It offers RAM options up to 16GB and, for the first time in this generation, a new RP1 I/O controller that connects to the main chip via a 4-lane PCI Express 2.0 bus to handle some of the I/O-related computational tasks. Due to the significant differences in both speed and performance, the Raspberry Pi 5 is the clear choice for our compute-intensive system.



Fig. 3.1. Raspberry Pi 5 microcomputer(Raspberry Pi, n.d.)

The optical sensor, Raspberry Pi Camera Module 3 Wide (*Raspberry Pi Camera Module 3 Standard NoIR Wide NoIR Wide*, 2024b), is essential for initiating our workflow and enabling further analysis. It serves as the sole source of input data for the computer vision model, making the quality and performance of the Sony IMX708 sensor, with 11.9 megapixels resolution, critically important for our system. Every setup and system is unique; with that in mind, this product's wide variant has a 120-degree diagonal field of view, a 102-degree horizontal field of view, and a 67-degree vertical field of view. This decision was made to give end users of the system more freedom to mount the unit at their convenience while still getting the most out of its wide field of view in all planes. The standard model was selected due to this system being mostly used during daylight hours.

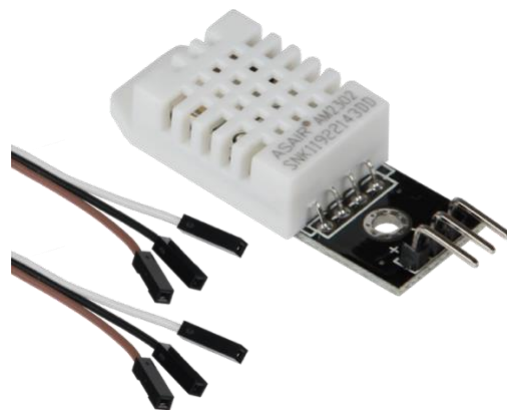


Fig. 3.2. Adafruit AM2302/DHT22 temperature and humidity sensor (LEMONA, n.d.)

Contextual information makes our system truly agentic. If our system relied solely on the initial input image, it would be a single point of failure, but some external factors could be accounted for by other sources, such as humidity and temperature. This is where the Adafruit AM2302 (Liu, n.d.) comes in. It acts as a safety net under certain conditions, preventing false positives and providing

more context for our agentic system. For solar panels, the most significant external factors are temperature and humidity, due to their sensitivity to weather conditions. Moreover, a temperature sensor can account for efficiency drops due to the physics of photovoltaic cells, with a 5% drop in power when the ambient temperature rises above 35 degrees Celsius. Instead of deducing that something went wrong, the agent could explain the power loss using the laws of physics rather than hallucinating the next-best answer. This sensor consists of three main parts. Capacitive humidity sensor for absorbing moisture, Thermal resistor for temperature changes, and to convert these analog signals to digital signals that the microcomputer can understand, and an 8-bit Microcontroller. With an operating range of -40 to 80 degrees Celsius and an affordable price, the AM2302 is a great choice for adding more environmental context to our system.

3.2. Electrical integration

To assemble the chosen hardware components into a unified edge-computing device, a centralized electrical system was designed inside the weatherproof enclosure. As shown in the electrical block diagram in Fig. 3.3, the Raspberry Pi 5 acts as the main processing unit, managing power distribution and data collection for all peripherals.

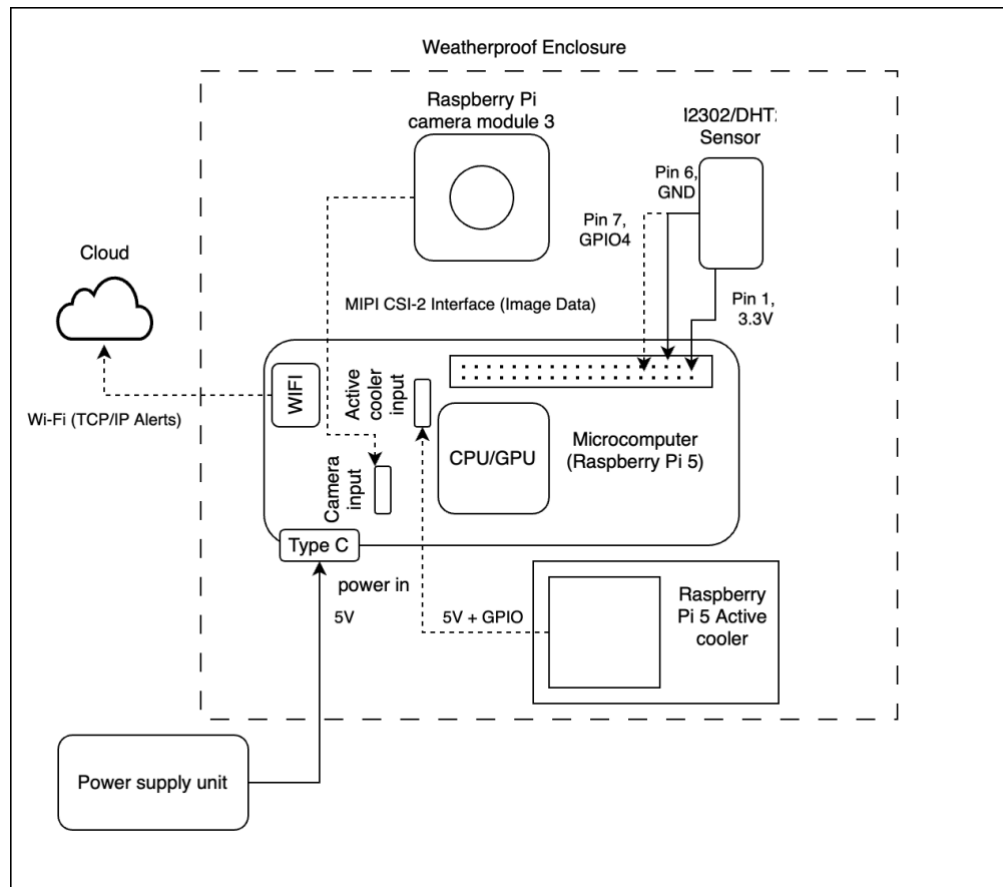


Fig. 3.3. Electrical block diagram

The system is supplied by an external 5V DC power source connected directly to the microcomputer's Type-C port, delivering the current needed to support both the board and its connected modules during peak AI inference activities. Thermal regulation is electrically managed through a dedicated active cooler input that provides 5V and PWM control straight from the board to keep temperatures at optimal levels. This active cooler has two parts; one is the passive heat dissipaters and the other is the active cooler's fan to push the heat away. It mounts natively on the Raspberry Pi 5's board.

Data acquisition depends on two main physical interfaces. The Raspberry Pi Camera Module 3 connects through a dedicated MIPI CSI-2 interface, enabling high-bandwidth, low-latency image data transfer directly to the CPU/GPU for computer vision tasks. In contrast, the Adafruit AM2302/DHT22 environmental sensor uses a low-bandwidth digital interface wired directly to the General-Purpose Input/Output (GPIO) header. To ensure safe signal logic, the sensor is powered via Pin 1 (3.3V) and grounded via Pin 6 (GND), with the digital data transmitted to Pin 7 (GPIO 4). Lastly, the system sends its processed data to the cloud and notification services using the microcomputer's built-in Wi-Fi module, completing the hardware-to-cloud communication pathway.

3.3. System flow and trigger logic

The execution logic begins with different types of triggers. The default method is the periodic trigger every hour, but the system can also be triggered manually via the solar sentinel interface. Additionally, it can be triggered infrequently but reliably using the Adafruit AM2302/DHT22 thermal and humidity sensor if temperatures exceed 35 degrees Celsius, as solar panel efficiency decreases by each degree above 35 degrees Celsius (Skoplaki & Palyvos, 2009). Conversely, freezing temperatures below 0 degrees Celsius can cause icing on top of the solar panels, blocking sunlight and adding structural weight that may damage some solar panel setups. Similarly, the humidity sensor registers saturation even when relative humidity exceeds 85%. Moisture on the solar panels leads to particulate cementation, which should be cleaned before it diminishes system efficiency (Ilse et al., 2019).

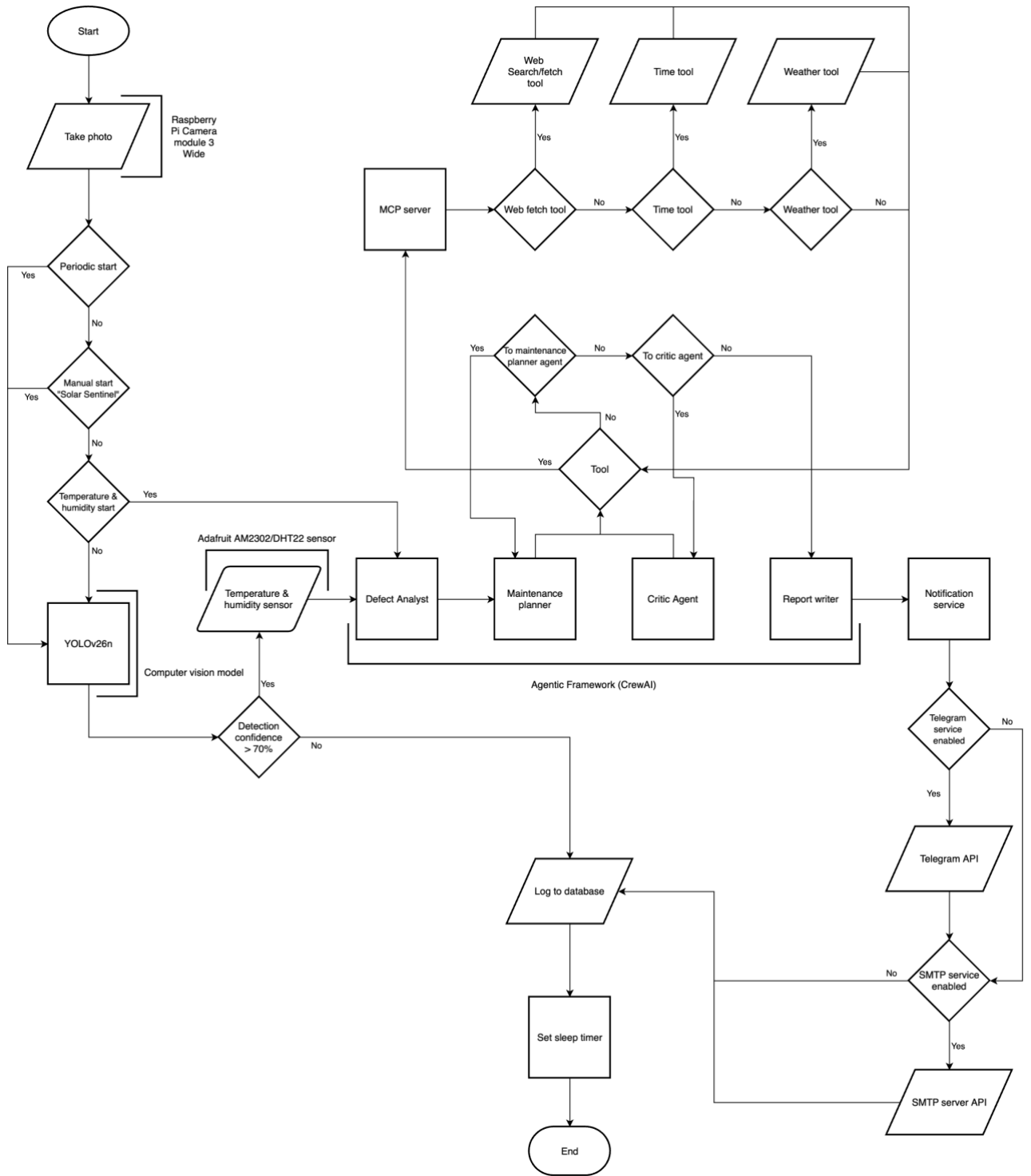


Fig. 3.4. Control algorithm

The entire control algorithm shown in Fig. 3.4 can be divided into three main parts. The “Entry trigger” part outlines the various ways the system can be triggered and the control algorithm can start. Secondly, the “Agentic workflow orchestration” stage includes the agentic workflow that serves as the comprehension layer, bringing “intelligence” to the system. This is a complex workflow with multiple agents that have access to multiple MCP tools to enhance capability. Later, the “Notification

service” informs the user via the user’s preferred method about the comprehensive report generated regarding the status of the solar panel system, and the system resets its trigger for the next run.

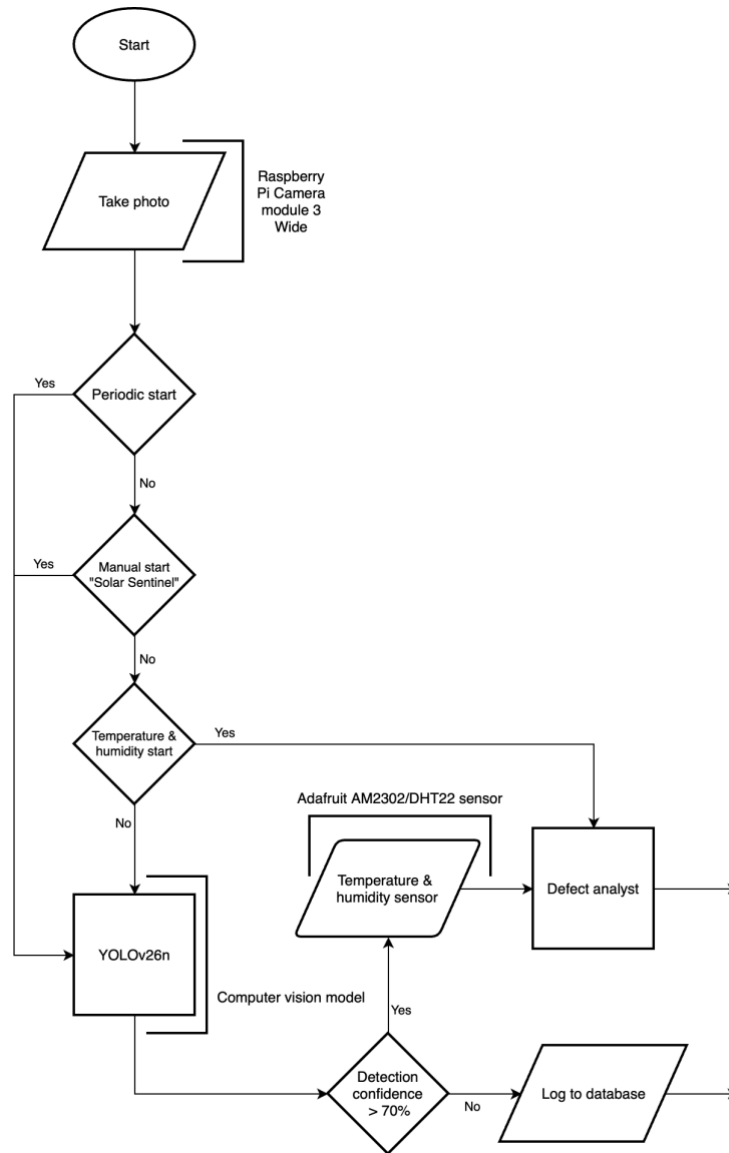


Fig. 3.5. Algorithmic flow part 1

Moreover, as shown in Fig. 3.5, the photo-taking process is completed after different trigger types. Then, depending on the trigger type, the computer vision model is activated. Whether something is detected or the environmental trigger activates the system, our agentic workflow begins with further comprehension and report generation, starting with the “Defect analyst” agent. However, if the detection confidence is below 70%, the captured image is logged to the database without agentic inference to reduce costs.

After the trigger by the YOLOv26n, image data, along with the temperature and humidity data, is then passed to the agentic workflow. This orchestration consists of 4 agents, 1 MCP server, and 4 tools connected. Essential aim of this orchestration is to mimic a professional solar panel maintenance

team. The systems team consists of 4 members (or agents). These agents are defect analyst, maintenance planner, critic agent, and a report writer. Each with their granular agent and task prompts, along with their tools, makes a personalized agentic system to help solve and maintain solar panel systems affordably.

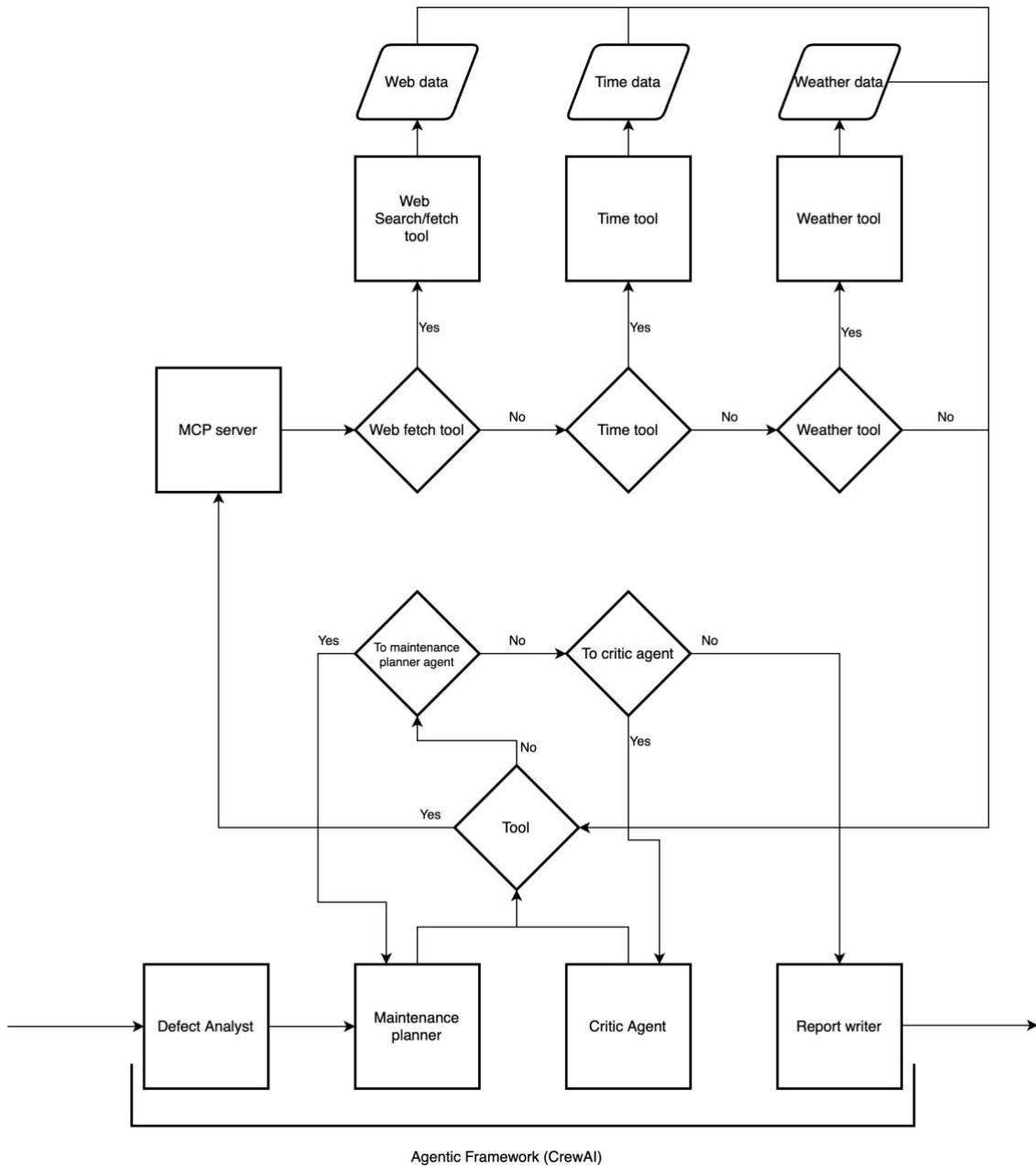


Fig. 3.6. Algorithmic flow part 2.

As shown in Fig. 3.6, the crew begins with defect analysis. The initial agent is responsible for utilizing its visual language model (VLM) capabilities to interpret what was detected in the image and understand the surrounding environment. This process confirms that the computer vision model's detection is not a false positive and helps clarify the system's underlying issue. Additionally, the maintenance planner agent provides further context by accessing tools like web search/fetch, time,

and weather forecast. These tools supply essential contextual information for our use case. For example, time data can inform sunlight and sunset patterns, weather forecasts can help prepare for future conditions, and the web search/fetch tool can be employed as a best practice search resource. Next, the critic agent comes into play. Its primary role is to serve as a quality assurance component, double-checking facts against the web and scrutinizing the claims made by the maintenance planner agent in pursuit of accuracy. It verifies everything based on a set of fixed logic rules and standards embedded in its prompt. Finally, the report writer agent compiles all these findings and data into a user-friendly, informative format to be later distributed to the end user via preferred methods.

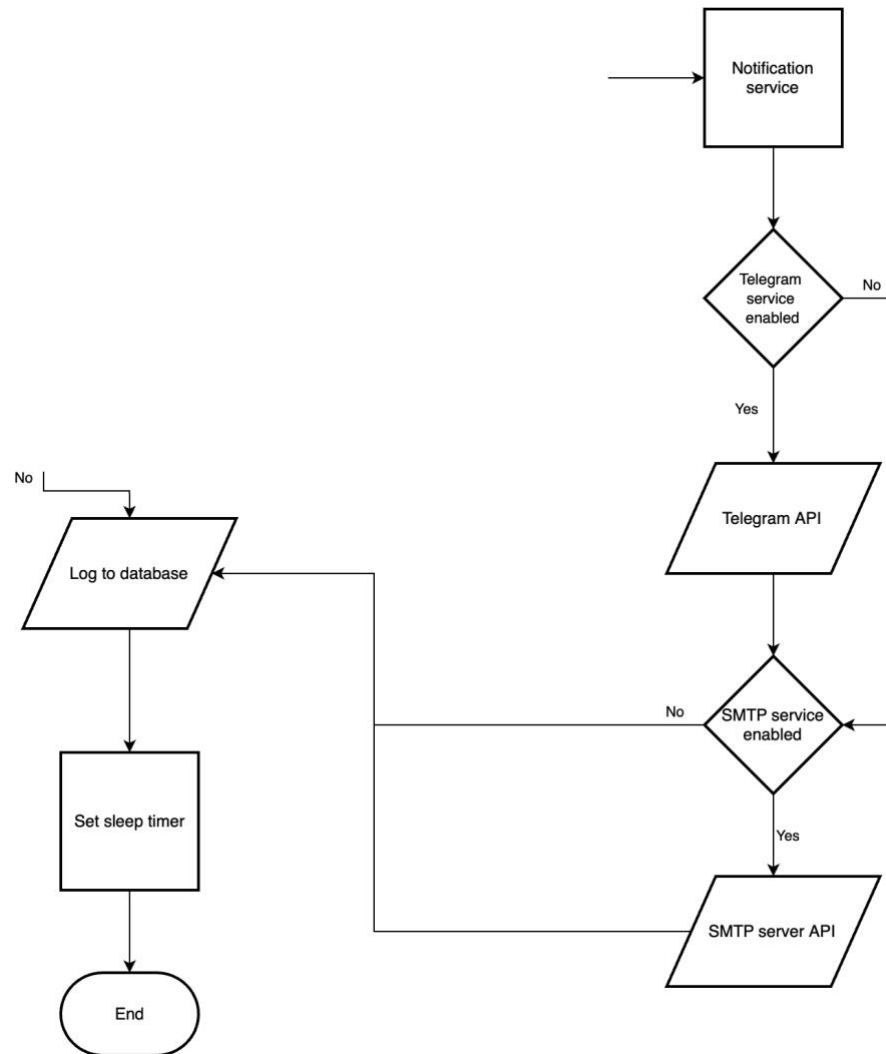


Fig. 3.7. Algorithmic flow part 3.

Then, the agentic workflow proceeds through the remaining agents, using the MCP tools they have access to. There are a few of them, such as the web search and fetch tool, time tool, and weather tool. Only “Maintenance planner” and “Critic Agent” have access to them by design. After the “Report writer” agent writes the report, it is passed to the notification service, which informs the user via two available methods. One is the Telegram service, which uses the official Telegram API. The other is the SMTP server, which uses the provided SMTP servers to send emails. After the enabled

notifications are sent, the actions and the report are logged to the system. Lastly, the sleep timer is updated based on conditions for the next iteration.

3.4. Solar Sentinel system

To bridge the gap between these two AI applications and serve as the backbone of the system, FastAPI serves as the central orchestration layer, passing the output of one step as input to another. It’s a high-performance, asynchronous framework built on the Asynchronous Server Gateway Interface (ASGI) standards, allowing it to efficiently process requests asynchronously. The frontend is a lightweight, simple page built with JavaScript, HTML, and CSS, without any frameworks. Since this page is hosted locally on the Raspberry Pi’s local network and no authentication is implemented, no frontend frameworks have been used. When run, the solar sentinel will be hosted on “http://localhost:8000” on the local network only. The code is hosted on GitHub: <https://github.com/p3rc1val/solar-sentinel>.

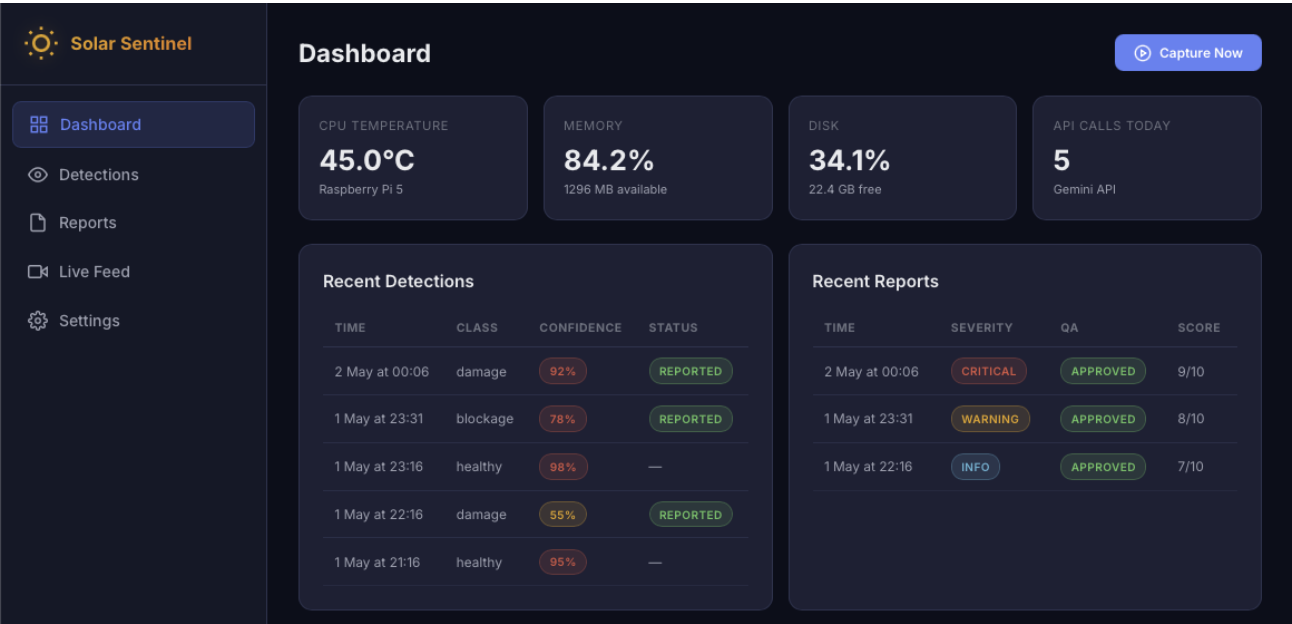


Fig. 3.8. Dashboard page of “Solar Sentinel”

Although it can later be set up to be reached from other networks via tailscale, the initial implementation keeps everything local and secure. When the link is launched, the user is greeted with an informative, concise dashboard that summarizes all the necessary information at a glance. This information includes basic system metrics such as CPU temperature, memory usage, disk usage, and today’s API calls, as shown in Fig. 3.8. Moreover, recent detections and reports are also presented as a means of quick access to the system’s history. Lastly, there is a “Capture Now” button that enables users to manually trigger their system whenever they want.

ID	TIME	CLASS	CONFIDENCE	PANEL	REPORT
1	2 May at 00:06	damage	92%	demo-panel-4	1
2	1 May at 23:31	blockage	78%	demo-panel-2	2
3	1 May at 23:16	healthy	98%	demo-panel-1	—
4	1 May at 22:16	damage	55%	demo-panel-3	3
5	1 May at 21:16	healthy	95%	demo-panel-2	—

Fig. 3.9. Detection history page of “Solar Sentinel”

The subsections in the left-hand menu provide easy navigation between different tabs. The detection history page displays all the triggered detection history. This information is stored in an SQLite database and shows all the relevant details about these previous runs, such as the time, classification, and confidence percentage.

ID	TIME	SEVERITY	URGENCY	QA SCORE	APPROVED
1	2 May at 00:06	CRITICAL	High	9/10	YES
2	1 May at 23:31	WARNING	Medium	8/10	YES
3	1 May at 22:16	INFO	Low	7/10	YES

Fig. 3.10. Reports page of “Solar Sentinel”

The reports section enables users to view each report created at a certain time. It is worth noting that not all runs pass the confidence threshold of 70% for the locally run YOLOv26n. Therefore, we have fewer reports than runs. Moreover, in Fig. 3.10, we can see the severity of each report, such as critical, warning, and info.

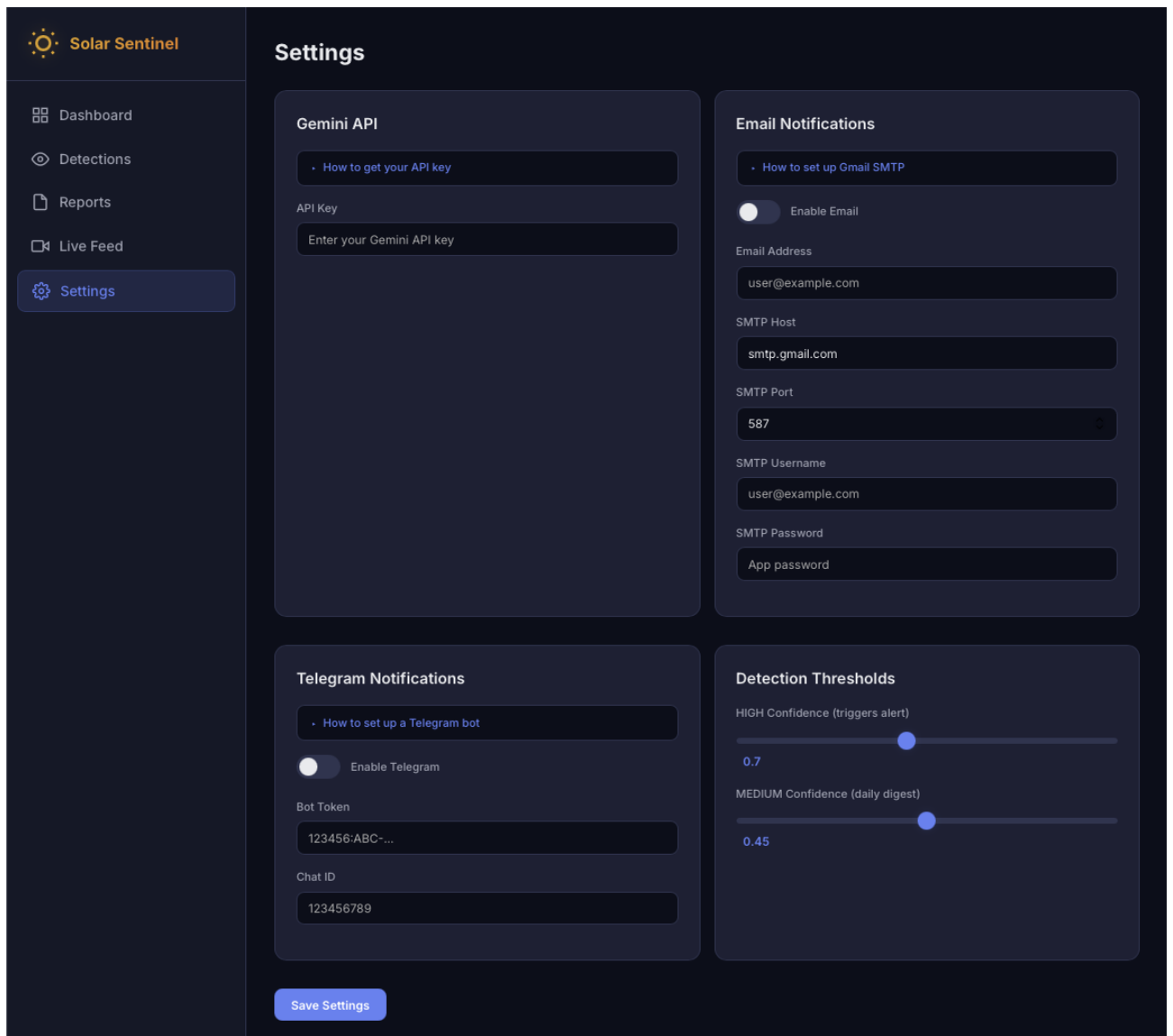


Fig. 3.11. Settings page of “Solar Sentinel”

The settings page shows all the customizations and controls. The first thing to note is the bring-your-own-key approach to requesting a Gemini API key. Then come the notification service parameters, such as secrets and the setup of the SMTP server and Telegram notification server. Lastly, the user can set the detection thresholds for high confidence, which triggers alerts, and for medium confidence, which triggers a daily digest, via an intuitive slider UI control.

4. SAFETY AND ENVIRONMENTAL IMPACT

The deployment of autonomous monitoring systems in residential settings requires a thorough evaluation of safety, privacy, and environmental standards. Although the technical effectiveness of the proposed solution is the main focus of this research, its real-world usability depends on its capacity to operate without risking users, infringing on data rights, or causing ecological harm. Unlike utility-scale solutions that work in controlled, isolated environments, a residential system operates near human activity and private property. Therefore, the system design must address these "soft" constraints as carefully as it does defect detection accuracy. This chapter covers safety protocols for the hardware, protections against algorithmic failure, legal regulations for data privacy, and the overall environmental impact of the device.

4.1. Work safety

The physical safety of the proposed system depends on electrical isolation and thermal management, especially because of the outdoor environment. The system uses a Raspberry Pi 5, which is efficient but generates significantly more heat than older models. When placed in a custom 3D-printed enclosure exposed to direct sunlight, internal temperatures can quickly exceed the safe limit of 85 °C. This could cause battery failure or thermal runaway. To prevent this, the enclosure includes passive ventilation channels and uses white, UV-resistant ASA (Acrylonitrile Styrene Acrylate) filament instead of standard PLA or ABS. ASA offers better weather resistance and reflects solar radiation, helping to keep the device within safe temperature limits without the need for energy-consuming active cooling fans. Additionally, the system runs on a low-voltage 5V DC setup (via USB-C), which is electrically isolated from the high-voltage DC strings of the solar panel array. This separation ensures that even if a major short circuit occurs in the camera unit, it won't energize the mounting frame or introduce dangerous voltages into the user's home network.

Operational safety also involves the inspection method. As mentioned in Chapter 1, commercial solutions often use crawling robots like the "SolarCleanser" or "HyCleaner," which traverse the PV modules. Although these methods are effective, they pose a risk of causing micro-cracks in the solar cells if the robot malfunctions or applies excessive pressure. The proposed solution employs a non-contact, tripod-mounted approach. By utilizing the high-resolution Sony IMX708 sensor in the Camera Module 3, the system captures detailed surface features from a distance of 1.5 to 2 meters. This distance eliminates the risk of physical damage to the panels during inspection. Additionally, the installation process is designed for "autonomous deployment," meaning homeowners won't need to climb onto roofs for manual checks. This practice helps reduce the number of fall-related injuries in residential maintenance.

4.2. AI safety and algorithmic reliability

The integration of Agentic AI introduces a layer of unpredictable behavior that must be managed to ensure reliability. LLMs like Gemini 3 Pro, which guide the system's decision-making, can produce "hallucinations" (Joshi, 2025). The model Gemini 2.5 Flash Lite performs well in terms of hallucination rates (*LLM Hallucination Leaderboard - a Hugging Face Space by Vectara*, n.d.), but it is still a smaller model compared to the selected Gemini 3 Pro, making it unsuitable for the current task. In maintenance scenarios, hallucinations could be costly. For example, the "Maintenance Planner Agent" might mistakenly interpret a shadow as a constant hotspot and advise the user to replace a functioning module. To prevent this, the system employs the CrewAI framework to enforce a strict validation process called "chain of thought." Before suggesting any action, a secondary "Critic Agent" reviews the recommendation against a fixed set of logical rules (e.g., "Do not recommend replacement unless efficiency loss exceeds 20%"). This careful process ensures that critical decisions are verified twice, reducing the risk of incorrect advice reaching the end-user.

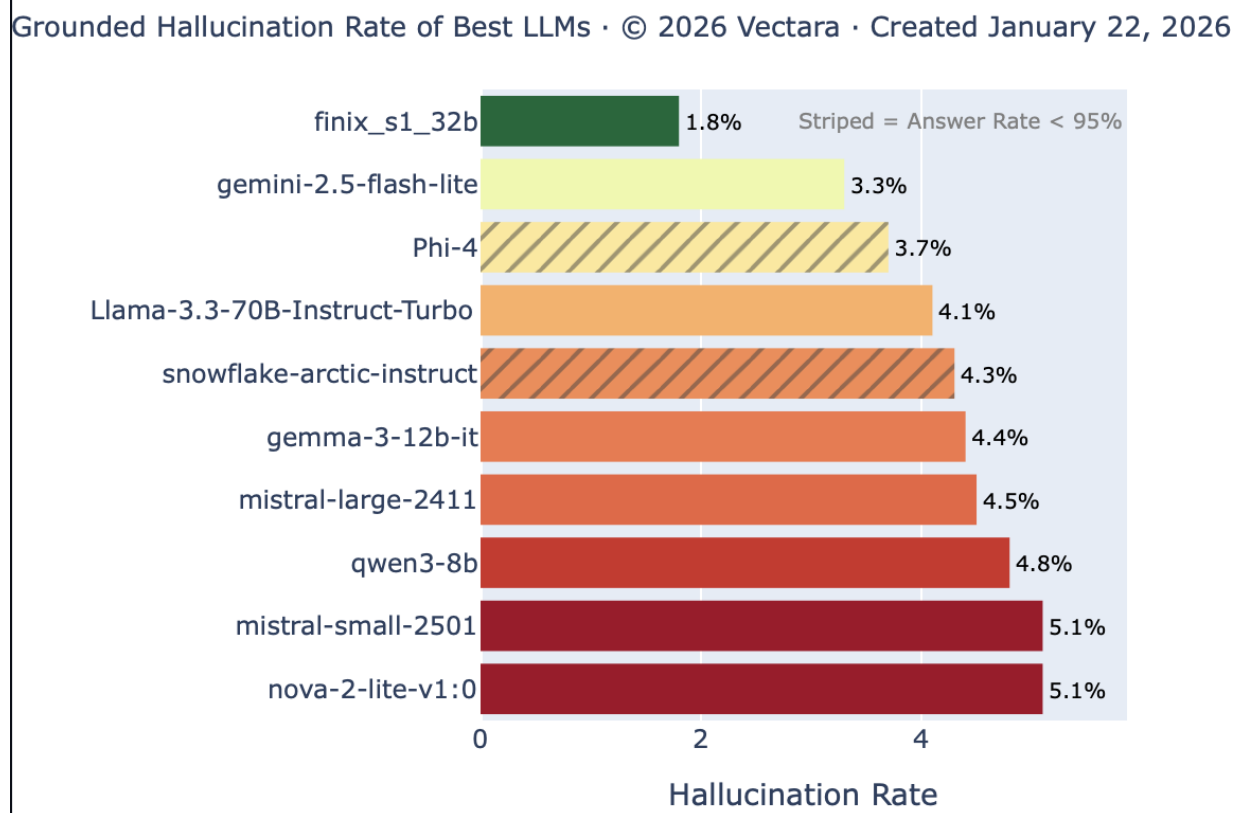


Fig. 4.1. Grounded hallucination rates of best LLM's by Vectara (*LLM Hallucination Leaderboard - a Hugging Face Space by Vectara*, n.d.)

Moreover, computer vision models like YOLO can have algorithmic bias, especially in unpredictable conditions. Research has shown that object detection models can display different error rates based on demographic characteristics if not properly assessed. While the main goal is to spot

defects on solar panels, the camera's wide view can capture reflections or people nearby. To handle this, the detection pipeline is carefully limited to the area where the solar array is located. Any objects found outside the defined region of interest are removed during the pre-processing stage. This not only speeds up the model's inference by narrowing the processing area but also stops the system from making judgments about non-target subjects, thus following the principles of fairness and purpose limitation in AI use.

4.3. Data privacy

In residential settings, privacy is a key legal requirement. The system will likely be installed in suburban areas where the camera's view might unintentionally cover neighbouring properties or public streets. To meet regulations like the General Data Protection Regulation (GDPR) in Europe, the system uses an edge-computing setup as a privacy shield. Unlike cloud-based cameras that stream video continuously to remote servers, the solution processes video feeds locally on the Raspberry Pi 5. The "Scan and Sleep" method ensures that the camera operates only during specific inspection times, such as five minutes each day. Raw video data stays on the local device; only selected, cropped images of confirmed defects are sent to the cloud for analysis. This method follows the idea of "Data Minimization," making sure the data collected is adequate, relevant, and limited to what is necessary for the maintenance task.

Additionally, the system uses "Privacy by Design" with software-based masking. During the initial setup, the user defines the four corners of the solar array. The software then adds a permanent black mask to all pixels outside this area before sending the image to the AI model. This ensures that even if the camera is aimed at a neighbor's window, the software cannot "see" or process that area. Data transmission is secured with industry-standard TLS 1.3 encryption when the system communicates with the LLM API. This helps prevent "man-in-the-middle" attacks that could intercept sensitive information about the home's energy system. By keeping most of the processing close to the source, the system shifts from surveillance to strict monitoring, matching technical function with legal privacy needs (Gao et al., 2021).

4.4. Environmental impact and sustainability

The environmental validity of the proposed system relies on a positive Net Energy Gain (E net), where the energy saved from detecting defects is greater than the energy used by the monitoring device. A regular Raspberry Pi 5 setup uses about 5 watts during active inference. However, with the "Scan and Sleep" logic, the device spends most of its time in a low-power state, using very little

energy. In contrast, a moderate layer of dirt on a 5kW residential solar array can decrease the output by 5% to 10%, which leads to a loss of about 250 to 500 kWh each year.

Table 4.1. Comparison of energy efficiency between cloud and edge AI architectures

Metric	Cloud SaaS AI (e.g., Server GPU)	Edge AI (e.g., Raspberry Pi/NPU)	Improvement Factor
Energy per Inference	~ 1 - 10 Watts	~ 1 - 10 milliwatts	~ 1000x
Latency	100 - 500 ms	< 10 - 20 ms	~ 10 - 50x
Data Breach Risk	High (Centralized)	Low (Localized)	N/A

From a lifecycle angle, the system design helps reduce electronic waste (e-waste). Using general-purpose, modular hardware like the Raspberry Pi extends the device's usefulness beyond one function. Unlike proprietary “black box” monitoring hubs that become outdated when the manufacturer stops support, the Raspberry Pi can be repurposed or recycled easily. The choice of materials for the enclosure matters too; using biodegradable PLA or recyclable PETG for 3D printing lessens dependence on new injection-molded plastics. Additionally, by using a small, quantized YOLOv11n model for inference, the system reduces the "carbon cost of compute." Large-scale cloud inference requires huge data centres that consume a lot of water and energy for cooling. By carrying out the heavy processing on the edge device using locally produced solar power, the system separates its operation from the carbon-heavy grid, providing a genuinely sustainable maintenance solution (Marpu et al., 2025).

5. ECONOMIC CALCULATION

This section provides an economic assessment of the autonomous solar panel defect detection system. To calculate the commercial viability of this system, various analyses were done, such as Capital Expenditure (CAPEX) for the initial expenses, Operational Expenditures (OPEX) for operating costs, Return on Investment (ROI) for comparative solutions, and last but not least, the Break-Even Point (BEP) is demonstrated to show the financial accessibility of this device for the target audience of residential owners.

5.1. Capital expenditure

The initial investment required to build the proposed system entirely consists of hardware components. These components are mostly Raspberry Pi-branded parts, such as the Raspberry Pi 5 microcomputer, the Camera Module 3 Wide, the active cooler, and the Adafruit AM2302/DHT22 thermal and humidity sensor. For the system, other materials' brands or origins don't really matter aside from the technical specifications. Those include the 128GB MicroSD card, the camera cable, and the custom 3D-printed enclosure. For local storage, it is important to have at least 128GB due to past attempts being recorded on the Raspberry Pi. However, it is still suggested to go with a high-performance variant, since the SD card is the only storage within the device that also hosts the operating system. It is essential for it to be reliable. Therefore, for our prototype, we went with the "Philips Ultra Speed A1 128GB" model (*Highlights Specifications*, 2020). As for the custom 3D-printed enclosure, since we are using ASA filament, the price for 1kg of filament in Lithuania is 14.90 EUR at the time of this thesis (*ASA 3D Filament 1.75 Mm | FILACRAFT*, n.d.) Assuming our enclosure is approximately 200 grams maximum, we can calculate the exact Bill of Materials for our system. Lastly, it is worth mentioning that due to global instability and hardware shortages caused by generative AI, prices are volatile and subject to change in every region.

Table 5.1. Bill of materials and their descriptions

Component	Description / Specification	Cost (EUR)
Microcomputer	Raspberry Pi 5 (8GB RAM)	129
Optical Sensor	Raspberry Pi Camera Module 3 (Wide)	45
Temperature & humidity sensor	DHT22/AM2302 Sensor	9
Thermal Management	Raspberry Pi Active Cooler	8
Local Storage	128 GB MicroSD Card	20
Connectivity	Raspberry Pi Camera Cable (200mm)	2
Enclosure	Custom 3D Printed ASA housing (~200g)	3
Total BOM		216

The bill of materials were gathered from official sources depending on the component. These sources include the authorized reseller of the Raspberry Pi in Lithuania (*RaspberryPi.Dk • Leading Raspberry Pi Shop in the Nordic/Baltic*, n.d.) for all the Raspberry Pi products, and for the temperature and humidity sensor, local providers provided the best offers, such as Lemona (LEMONA, n.d.-b).

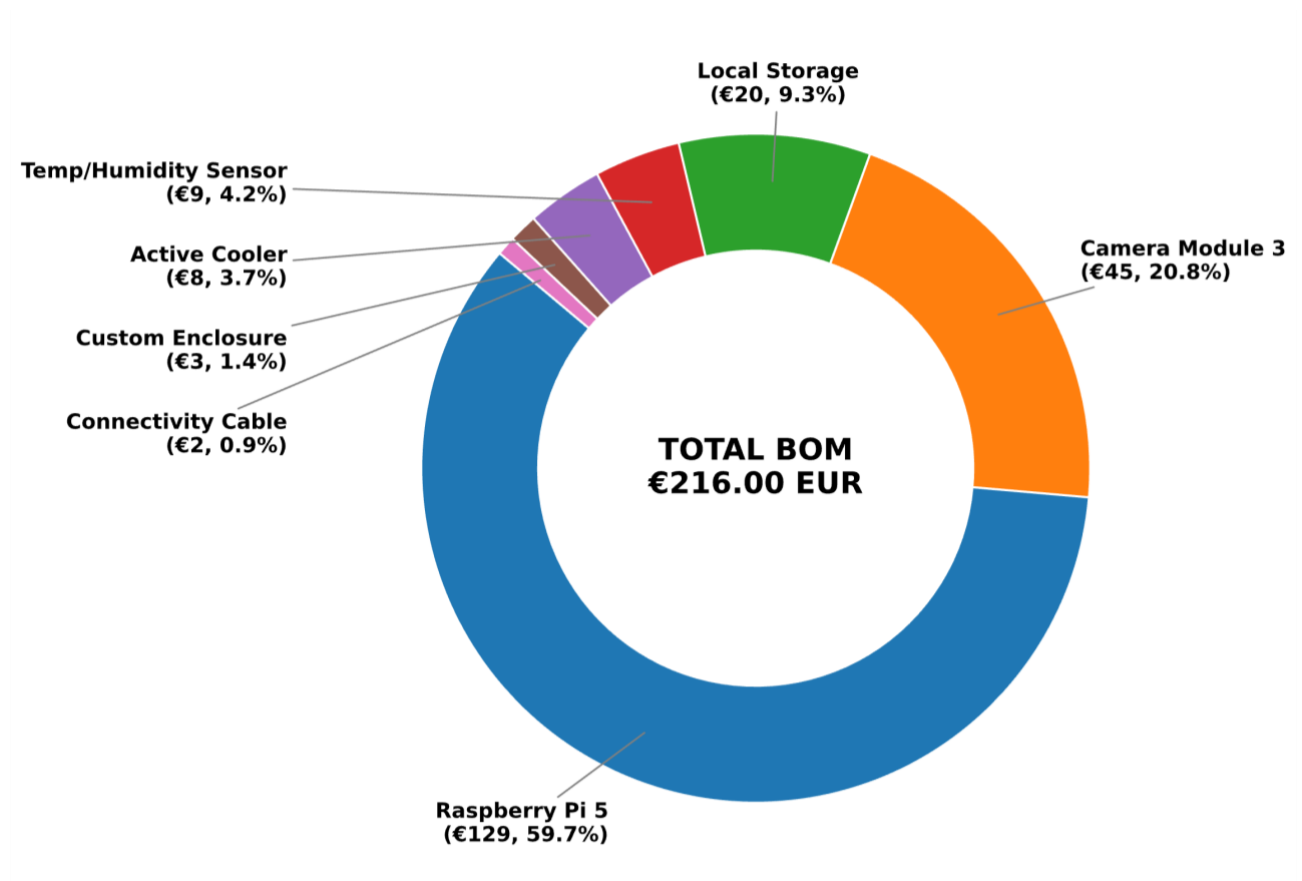


Fig. 5.1. BOM distribution displayed in a pie chart

Figure 5.1 clearly shows the BOM distribution by component. Naturally, as the most important part of the system, the microcomputer accounts for 59.7% of the total cost. Next comes the Camera Module 3 Wide, featuring high-definition imaging, a premium Sony IMX708 sensor, and autofocus. Then comes local storage, followed by the temperature and humidity sensor. Last but not least, the least expensive components are the active cooler, custom enclosure, and connectivity cable.

The BOM alone does not cover the entire variable manufacturing cost per unit. The cost of outsourced 3D printing for the custom enclosure, as well as manual assembly and software configuration, must be included to assess commercial viability accurately. Industrial 3D printing costs vary by provider; for example, Lithuanian suppliers like “GoMy3D” charge around 60 EUR per enclosure (*GoMy3D - High Quality 3D Printing Services for Custom Prototypes*, n.d.). Although this

is a small-batch price that would decrease with larger quantities, it provides a conservative estimate for initial production. After component fabrication, manual hardware assembly and OS flashing take up to 2 hours per unit. With an average monthly salary of 2000 EUR for an electronics technician in Lithuania, the hourly rate is 12.50 EUR, leading to a labor cost of 25 EUR per device for assembly. Thus, adding the 216 EUR BOM, 60 EUR printing fee, and 25 EUR labor cost results in a total direct variable manufacturing cost of exactly 301 EUR per device.

5.2. Operational expenditure

This system was designed with operational costs in mind at every decision point. To ensure self-reliance and long-term sustainability, every decision was made to reduce operational costs. From the smart trigger that only fires up based on the delay provided by the owner, to the edge device architecture, operational costs are kept very low. Moreover, the Agentic AI workflow layer has a smart fallback mechanism that uses all Gemini models within the generous free tier (*Rate Limits | Gemini API | Google AI for Developers*, n.d.), making the system even more cost-efficient for third-party LLM calls. This feature can be updated to cover other providers, offering even more variety in free LLM usage. This is made possible by the locally run YOLOv26n model, which acts as a filter, allowing only selected warnings to pass through. Lastly, the system's energy costs are drawn only from the Raspberry Pi's maximum 5-watt power draw. Although it is hard to calculate the exact usage due to the owner-adjustable sleep cycle, it is safe to say the electricity consumption is negligible for a system this capable.

A realistic commercial deployment model includes essential recurring costs beyond basic electricity and API fees. Hardware components that endure outdoor operation and thermal stress, such as the active cooling fan and the MicroSD card responsible for ongoing OS read/write cycles, have predictable replacement needs. The fan and MicroSD card typically need replacement approximately every 2.5 years. To accurately represent this within the financial analysis, an annual hardware amortization cost is combined with an estimated requirement of one maintenance specialist visit every three years. Taking these factors into consideration, the system's total operational expenditure (OPEX) is quantified at exactly 19.53 EUR annually.

5.3. Opportunity costs and return on investment

To put all the expenses into perspective, it is helpful to compare them with the potential financial losses from neglected systems and the cost of third-party maintenance services. According to 2026 solar panel maintenance market data, a professional residential solar panel cleaning and inspection in Europe costs an average of 125 to 150 EUR per session (Matar & An, 2024). Assuming

that a homeowner relies on an action-based maintenance approach, unchecked soiling and micro-defects can reduce system efficiency by an average of 10% (Chala et al., 2025). For example, a 5 kWp residential setup producing roughly 5000 kWh annually would lose 500 kWh of energy. To convert this lost energy into euros, we can make a simple calculation using the 0.22 EUR per kWh price tag that the popular Lithuanian energy provider Ignitis guarantees (*Visuomeninis Elektros Tiekimas | Ignitis*, n.d.). This translates to 110 EUR per year in lost energy generation alone.

$$ROI = \left(\frac{\text{Annual Gross Savings} - \text{Annual OPEX}}{\text{Total CAPEX}} \right) \times 100 \quad (5.1)$$

$$ROI = \left(\frac{110 - 19.53}{475} \right) \times 100 = \left(\frac{90.47}{475} \right) \times 100 \approx 19.05\%$$

The return on investment (ROI) is mathematically determined by evaluating this recovered energy value against the system's commercial retail price. By taking the 110 EUR gross annual energy savings and subtracting the 19.53 EUR annual OPEX, the net annual benefit is established. This figure is then divided by the total customer capital expenditure (CAPEX) of 475 EUR and multiplied by 100. As demonstrated by ROI equation (Blank & Tarquin, 2024), this yields an annual ROI of 19.05%, offering a highly competitive yield compared to traditional financial instruments.

5.4. Commercial break-even point

To accurately assess the commercial feasibility of producing this monitoring system, conducting a BEP (break-even point) analysis is crucial. The break-even point indicates the sales volume needed for total revenues to exactly cover total costs. To determine this, a profitable selling price must be set that covers both variable costs and fixed overheads. For this system, the direct variable cost per unit (v) is 301 EUR, which includes materials and assembly labour. A retail price (P) of 450 EUR is suggested to achieve a gross profit margin of approximately 49.5%, representing a markup over variable costs. Additionally, annual fixed costs (f) are estimated at 6,000 EUR.

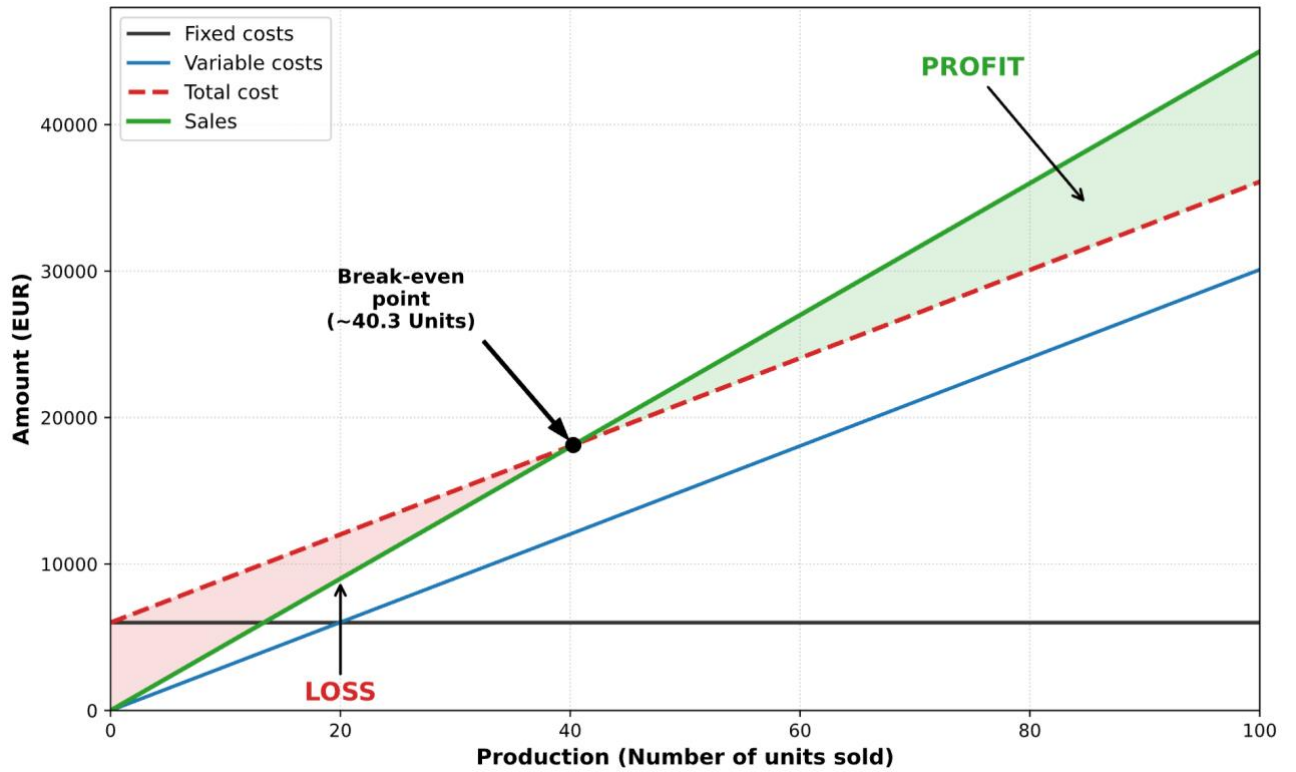


Fig. 5.2. Commercial BEP analysis.

Since fixed costs do not change with production volume, this amount covers recurring expenses such as marketing, operational costs, equipment depreciation, and sales costs. Using the standard break-even formula, the required annual sales volume can be calculated.

$$BEP = \frac{f}{P - v} \quad (5.2)$$

$$BEP = \frac{6000}{450 - 301} = \frac{6000}{149} \approx 40.27 \text{ units.}$$

Based on these parameters, the enterprise must manufacture and sell approximately 41 units annually to break even. Every subsequent unit sold beyond this threshold generates 149 EUR in gross profit, indicating a highly accessible commercialization path.

5.5. Payback period analysis

From the residential owner's perspective, the financial attractiveness of the system is best measured by the payback period. This metric identifies the exact timeframe required for the device to generate sufficient energy savings to recover the initial purchase price. Utilizing the established retail price, the total capital expenditure (CAPEX) for the customer is 475 EUR, which includes the 450 EUR hardware cost and a 25 EUR professional installation fee. The annual net savings are calculated at 90.47 EUR, derived from the recovered energy value (110 EUR) minus the system's

annual operational expenditure (19.53 EUR). The payback period (A) is determined by dividing the total investment by these annual savings:

$$A = \frac{\text{Total CAPEX}}{\text{Annual Net Savings}} \quad (5.3)$$

$$A = \frac{\text{Total CAPEX}}{\text{Annual Net Savings}} = \frac{475}{90.47} \approx 5.25 \text{ years}$$

The calculation shows that the homeowner will fully recover their initial investment by the first quarter of the sixth year of operation. Given that the hardware is engineered for long-term outdoor durability, the remaining 10-year projected lifespan will yield pure financial profit for the user. This strongly validates the system's economic value within the residential solar maintenance market.

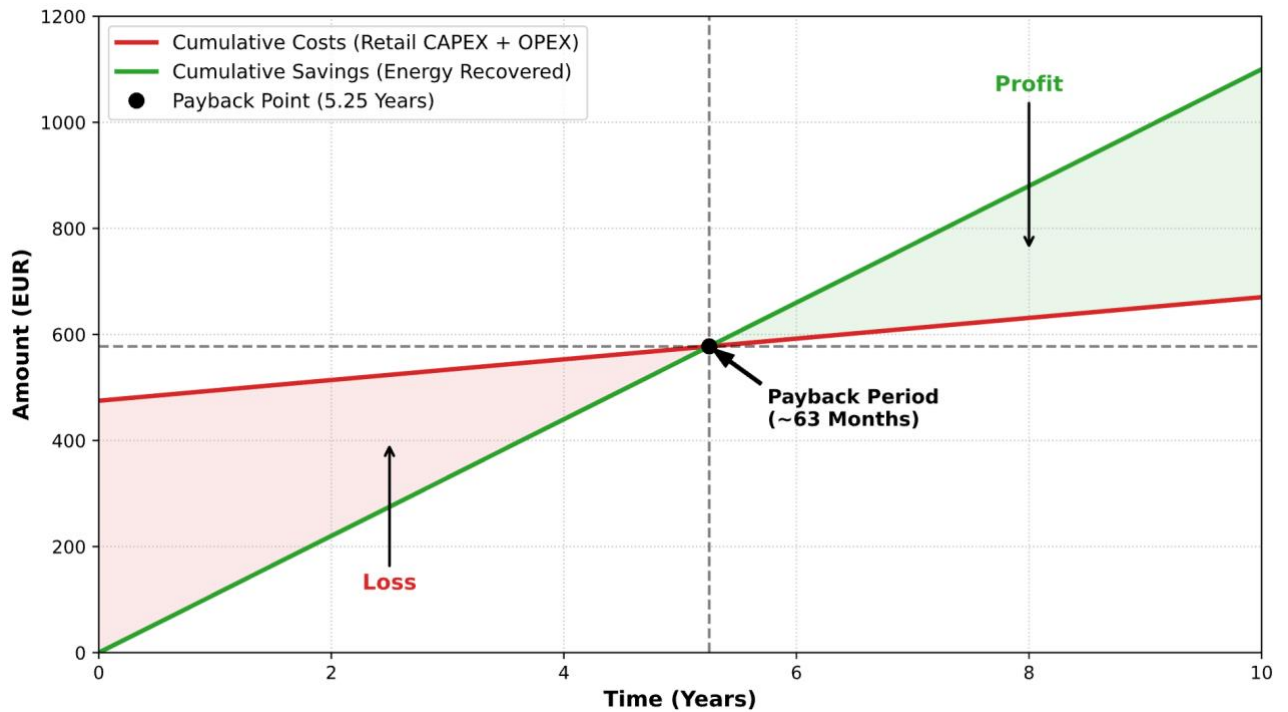


Fig. 5.3. Payback period analysis.

According to the payback analysis for the 10-year plan, the initial customer investment is fully recovered by the first quarter of the sixth year of operation, and the remaining years fall within the net profit zone. Although the system's components can operate for more than 10 years under stress, the chart shows that even if the system lasts for 10 years, a significant portion of that period falls within the net profit zone for the residential owner.

6. CONCLUSIONS

The popularity of sustainable energy resources comes with its own problems. The majority of these problems are solved for large-scale photovoltaic setups that produce gigawatts of power, due to the simple math of economies of scale, which make those massive setups the most efficient when you calculate the amount. However, residential installations, also referred to as “Rooftop solar photovoltaics,” account for 40% of the global solar photovoltaics (Joshi et al., 2021). Therefore, a significant portion of the market virtually doesn’t have a solution for an efficient, AI-powered, proactive maintenance system that doesn’t require a high initial investment or recurring subscription models. This thesis successfully addresses this market gap by proposing and validating an autonomous, cost-effective, and pragmatic monitoring system custom-designed for small-scale residential solar installations.

With the help of efficiency measures like edge computing powered by the Raspberry Pi 5 8GB, a high-resolution Raspberry Pi Camera Module 3, and an Adafruit AM2302/DHT22 temperature and humidity sensor, the system can detect discrepancies on its own via a custom-trained YOLOv26n model in a self-reliant fashion. Edge computing can be utilized in the system due to the “Scan and Sleep” architecture we have chosen. Based on a user-given period, it ensures that only minimally required computation is done on the microcomputer. The only dependencies come into play after the system detects and classifies the discrepancy and needs to run the Multi-agentic system. Even then, the smart fallback mechanisms that keep the LLM usage in the free tier of Gemini initially and any given LLM provider in the future ensure the system can maintain itself with negligible operational costs. This smart fallback system works thanks to the bring-your-own-key architecture style we have picked, and it starts from the best available model, in our case Gemini 3.1 Pro, then falls back to the next best one until the models are exhausted and goes to the next provider or awaits the availability of the next LLM call. All of this computation is managed thermally by the Raspberry Pi 5 active cooler. The main complexity and the greatest value of this system comes from its modern agentic approach that enables proactive maintenance rather than costly reactive maintenance, which is a total paradigm shift that protects the investments of residential solar panel owners.

Furthermore, the architecture enables safe local storage of data, and the bring-your-own-key approach offers endless possibilities when it comes to LLM choices. Moreover, edge computing enables safe and private processing of potentially sensitive image data, and LLM calls go through de-anonymization and software-based masking techniques to ensure the system is private by design. It also fully aligns with data minimization principles and regulations like GDPR. To fully have an enclosed and self-reliant edge compute device, open-source generative AI models should become more efficient and optimized for computational constraints. Although this doesn’t mean this system

couldn't have included a very small LLM, the free tier of popular providers makes cutting-edge models extremely accessible, and in order to have reliable analysis and comprehension, today's LLMs with fewer than 15 billion parameters, which barely could be run by the Raspberry Pi 5, cannot compete with the "intelligence" of a model that utilizes hundreds of billions of parameters.

Economically, the system is highly viable not only because of its low bill of materials of 216 EUR, low initial capital expenditures, and negligible operational costs, but also because of its strong return on investment of 19.05%, a commercial break-even point of just around 40.3 units, and a payback period of 5.25 years. Gaining about 10% of net energy production back in solar panel installations can justify further investments into sustainable energy.

Finally, this research demonstrates that by combining state-of-the-art agentic AI with highly efficient computer vision models that can run on edge hardware like YOLOv26n, professional-grade proactive maintenance can be scaled down to residential solar panel installations. Consequently, making sustainable energy resources even more accessible and preferable to traditional resources.

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THE ANNEXES

[illegible]

Format	Zone	Position	Tag	Title	Number	Note
				<u>Documentation</u>		
A1			MERS BM 26 2165 01 02 00 AD	Assembly drawing	1	
				<u>Parts</u>		
A4		4	MERS BM 26 2165 01 01 04	Fan Cover	1	
				<u>Standart components</u>		
		5		ISO 7045 M3x14 mm	2	
		6		<u>Spring</u>	2	
		7		ISO 7045 M2x12 mm	3	
				<u>Other Parts</u>		
		1		Heat dissapator	1	
		2		Motor blade	1	
		3		Motor	1	
MERS				Document type Specifications	Status of the document Education	
Owner VG TU TDIfu-22			Prepared by Baha Celik	Title Integration of Computer Vision and Natural Language Processing for Autonomous Solar Panel Defect Detection and Classification	Tag MERS BM 26 2165 01 02 00 SP	
			Approved by M. Makulavičius		Issue A	Date 2026 05 12
					Lang. Eng.	Page 1/2